

AMI — Figma Handoff Document (EN)

Exhaustive feature inventory for UI/UX rewrite

AMI Team

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Chapter 1

Preamble

Application: AMI — Assistance Missions Interprètes (Interpreter Missions Assistance) **Stack:** Flutter Web (frontend) + PHP (REST API on Dolibarr database) **Git version:** b75ebc9 **Generation date:** 2026-05-31

1.1 Purpose of this document

This document **exhaustively describes the existing features** of every page, screen, dialog and modal of the AMI application. It is intended for a **Figma** designer tasked with producing a **new UI/UX** while preserving **100% of the current business behavior**.

1.2 How to read this document

For each page or dialog, a standardized sheet is provided:

1. **Route / access** — how the user reaches the screen (URL, menu, UI action), role conditions.
2. **Business purpose** — why the screen exists.
3. **API endpoints** — PHP files called.
4. **Layout & zones** — current visual breakdown.
5. **Data displayed** — table column-by-column / field-by-field.
6. **User actions** — table action × trigger × precondition × effect × feedback.
7. **States & edge cases** — loading, empty, error, insufficient rights.
8. **Important business rules** — invariants to preserve.
9. **UI dependencies** — shared widgets, parent dialogs.
10. **Notes for the designer** — non-negotiable behaviors, Dolibarr-type constraints.

Each key behavior is cited with **file:line** as a Markdown link to the source code. Any screenshots (screenshots/<slug>.png) are only used as **visual reference**, never as specification: the textual description remains the source of truth.

1.3 Cross-cutting conventions and constraints

- **Dolibarr-type information density:** compact search bars on a single line, reduced font, strict column alignment, little superfluous vertical spacing.
- **Dolibarr-type expected responsiveness:** immediate interaction with the table without waiting for loading to finish (pagination/progressive scroll).
- **Text selection:** all dialogs use `showDialog` wrapped in `SelectionArea` (see [selectable_dialog.dart](#)) — text must remain selectable.
- **Roles:** admin, manager (mission management), interpreter. See access matrix in the *Navigation architecture* section.
- **Business colors:**

- Quote statuses: draft (gray), sent (blue), accepted (green), rejected (red), expired (orange), accepted_converted (purple).
- Invoice statuses: draft (FR: brouillon) (gray), validated (blue), paid (green), cancelled (red).
- **Currency format:** fr_FR locale, € symbol, 2 decimals.
- **Date format:** YYYY-MM-DD for storage, dd/MM/yyyy for display depending on context.

Chapter 2

1. Common screens and architecture

2.1 Home (Hub) — [lib/screens/home_page.dart](#)

Route / access: / (default route if `AuthManager.isLogged`) or direct access `Navigator.pushReplacementNamed(context, '/login')` after logout

Business purpose: Centralized hub for module access (Interpreters, Missions, Billing, Third Parties, Admin, Export) based on user rights.

API endpoints: None (orchestration only, endpoints delegated to child pages)

2.1.1 Layout & zones

- **AppBar:** Logo + title “AMI - Assistance missions interprètes”
- **TabBar:** Contextual tabs based on rights (`canManageInterpreters`, `canManageMissions`, `isAdmin`)
- **Desktop:** Tabs on one line (flex 4), user name “Welcome [USER]” (flex 5), logout btn
- **Mobile:** Logo + title + logout on row 1, full-width tabs under AppBar
- **Body:** `TabBarView` with child pages
- **Footer:** `BrandFooter`

2.1.2 Data displayed

Field	Source	Format	Sort/Filter	Notes
Logo	<code>assets/logo.png</code>	34×34 px image	—	“RF” container fallback if asset missing
Title	Literal string	Text	—	Desktop: “AMI - Assistance missions interprètes”, Mobile: “AMI”
Welcome	<code>AuthManager.userFullName</code>	First Name Last Name “LastName” text	—	Desktop only, format “\${user['prenom']} \${user['nom']}”

Field	Source	Format	Sort/Filter	Notes
Tabs	Routes/pages based on rights	Tab label text	—	Interpreters, Missions, Billing, Third Parties, Admin (if admin)

2.1.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Click tab	Touch/click tab label	—	Navigate to child page (TabBarView)	Tab highlighted, page loaded
Click Logout	Click logout button (logout icon)	—	<code>AuthManager.logout()</code> + <code>pushReplacementNamed('/login')</code>	Return to LoginPage

2.1.4 States & edge cases

- No tab available → empty display (rare case, user without rights)
- Screen resize → responsive layout (AppBar adapts to isMobile)
- Tab closed/reloaded → TabController keeps current tab

2.1.5 Important business rules

- Displayed tabs depend on `UserRights`: each access condition validated at runtime
 - Interpreters: `canManageInterpreters || isAdmin`
 - Missions/Billing/Third Parties: `canManageMissions || isAdmin`
 - Admin: `isAdmin` only
- Custom TabBar indicator: light background + colored border-bottom (primary)
- Logo fallback if image missing: border + “RF” text

2.1.6 UI dependencies

- `InterpretersPage`, `MissionsTablePage`, `BillingPage`, `RequestersManagementPage`, `AdminPage`, `CompanyInfoPage`, `ExportPage`
- `AuthManager.userRights`, `AuthManager.userFullName`
- `BrandFooter`

2.1.7 Notes for the designer (to preserve)

- TabBar indicator: `BoxDecoration` with primary color + 2.5px border-bottom
- AppBar: 68px height, elevation 0 (flat design)
- Responsive helper: 600px breakpoint for isMobile
- Hero gradient used on login, not here

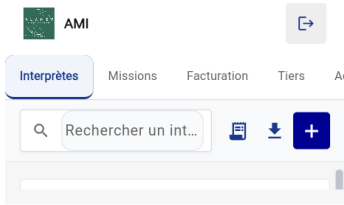


Figure 2.1: home

2.2 Administration — [lib/screens/admin_page.dart](#)

Route / access: /admin (Tab in HomePage if isAdmin())

Business purpose: Complete user management (CRUD): search, toggle rights, modification, deletion.

API endpoints: - GET api/admin/get_users.php — retrieve list + summary

- POST api/admin/update_user.php — modify user + rights

- POST api/admin/delete_user.php — delete user

2.2.1 Layout & zones

- **Title banner:** "Administration" in primary blue, "Company info" button (to /company-info)
- **Search area:** TextField with search icon, placeholder "Search a user..."
- **Table (desktop):** Primary color header, 5+4+4+4+4 columns (flex), alternating row zebra (white/light gray)
- **List (mobile):** Stacked cards, user title + email, stacked switches under divider
- **Table columns:** First Last Name | Interpreters (switch) | Interpreter Mgr. (switch) | Missions (switch) | Admin (switch) | Actions (Edit/Delete icons)

2.2.2 Data displayed

Field	Source	Format	Sort/Filter	Notes
Full name	DB <code>concat(firstname, lastname)</code>	Text	Full-text search	Main display result
Email	DB email	Text	Full-text search	Below name in gray
Interpreter	DB flag <code>is_interpreter</code>	Boolean switch	—	Real-time enable/disable
Interpreter Mgr.	DB flag <code>can_manage_interpreters</code>	Boolean switch	—	Real-time enable/disable
Missions	DB flag <code>can_manage_missions</code>	Boolean switch	—	Real-time enable/disable
Admin	DB flag <code>is_admin</code>	Boolean switch	—	Real-time enable/disable
Status	DB <code>statut</code>	"Active"/"Inactive" label	—	Computed from DB, not editable in UI

2.2.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Type search	Input in TextField	—	Update filtered list <code>_filtered</code> (full-text on full-name+email+username)	List re-filters in real time
Toggle right switch	Click switch (Interpreter/Mgr/Missions/Admin)	—	Call <code>_toggleRight()</code> → POST API <code>update_user</code>	Switch UI changes immediately, snackbar confirmation
Click Edit (pencil)	Click edit icon	—	User edit modal dialog	Dialog <code>_showEditDialog()</code> displays
Click Delete (trash)	Click delete icon	—	Confirmation dialog	Dialog <code>_confirmDelete()</code> displays
Click "Company info"	Click FilledButton	—	Navigate to <code>/company-info</code>	CompanyInfoPage loaded

2.2.4 States & edge cases

- Initial load → centered `CircularProgressIndicator`
- API error → red error message + snackbar
- No user found → "No user found" text
- Right toggle fails → error snackbar, data reload on fail
- Deletion fails → error snackbar, data reload on fail
- Mobile screen → list-cards layout instead of table

2.2.5 Important business rules

- Real-time search filters on `fullname`, `email`, `username` (lowercase)
- Rights are independent flags (a user can be admin + interpreter)
- Edit and Delete require explicit confirmations (dialogs)
- Right changes are immediate (optimistic UI), reload on error
- Status column displayed but not editable (info only)

2.2.6 UI dependencies

- `AdminService.getUsers()`, `AdminService.updateUser()`, `AdminService.deleteUser()`
- `UserModel` (class)
- `ResponsiveHelper.isMobile()`
- Custom dialogs via `showDialog` (wrapped in `SelectionArea`)

2.2.7 Notes for the designer (to preserve)

- “Full Name” column `flex=5`, other rights `flex=3`, Actions fixed width=96px
- Switch scale 0.85 (reduced)
- Table header: primary background + 13px bold white text
- Alternating zebra: white/#F8FAFC
- Edit icon primary blue, delete red #DC2626

2.2.8 Child dialogs

2.2.8.1 Dialog: User edit

Opening: Click Edit button on user row

Content: 4 text fields (Full Name, Email, Login, Password)

Buttons: Cancel, Save

Effect: POST `api/admin/update_user.php` with data, return snackbar + list reload

2.2.8.2 Dialog: Deletion confirmation

Opening: Click Delete button on user row

Content: “Delete [fullname]?”

Buttons: Cancel (gray), Delete (red)

Effect: POST `api/admin/delete_user.php`, return snackbar + list reload

2.3 Data export — [lib/screens/export_page.dart](#)

Route / access: `/export` (accessible if `isAdmin()`)

Business purpose: Export business data (Interpreters, Users, Missions) as CSV for third parties/reporting.

AMI - Assistance missions interprètes		Interprètes	Missions	Facturation	Tiers	Admin	Bienvenue Anna FALL
Administration							Infos entreprise
Rechercher un utilisateur...							
Nom Prénom	Interprètes	Gest. Interprètes	Missions	Admin	Actions		
MAJID	☒	☐	☒	☐	✎	✖	
MISBA	☒	☐	☒	☐	✎	✖	
NELLY	☒	☐	☒	☐	✎	✖	
IBRAHIM (De la part de TESFIT)	☒	☐	☒	☐	✎	✖	
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Adam Abdoulay Abakar	☒	☐	☐	☐	✎	✖	

Figure 2.2: admin-users

API endpoints: - GET `api/admin/get_users.php` — user list for export

- GET `api/get_missions_datatable.php?q=...&page=...&pageSize=...` — paginated mission loading

- GET `api/getMissionsDatatableAll()` — complete export of all missions

- GET `api/interpreter_service.getInterpreters()` — retrieve interpreters

2.3.1 Layout & zones

- **Status message:** Displays export state (loading..., done, error)
- **Wrap of export buttons:** 5 main buttons (Interpreters CSV, Users CSV, Show Missions, Export Table CSV, Export All CSV)
- **Mission controls area:** Search field + page size dropdown + pagination buttons + page/total display
- **Missions DataTable:** Quote Ref, Client, Interpreter, Start, Product, Product Id columns
- **User tip:** "Tip: use Chrome on desktop to download"
- **Footer:** BrandFooter

2.3.2 Data displayed

Field	Source	Format	Sort/Filter	Notes
Export status	_message property	Descriptive text	—	"Exporting interpreters...", "Missions exported", "Export error..."
Missions (DataTable)	Paginated API	Text columns	Full-text search	rowid, reference_devis, client_name, interpreter_name, debutmission_iso, produit_ref, id_produit_service
Page size	Fixed dropdown	[10, 25, 50, 100, 200]	—	Default 50, modifiable without reload
Paging	_page, _total state	"Page 1 • Total 156"	—	Current display / sum

2.3.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Export Interpreters CSV	Click "Export Interpreters (CSV)" button	web only	Call <code>_exportInterpretersCsv()</code> → generate CSV → download	Message "Exporting interpreters..." then "Interpreters exported"
Export Users CSV	Click "Export Users (CSV)" button	web only	Call <code>_exportUsersCsv()</code> → GET <code>admin/get_users</code> → CSV → download	Message "Exporting users..." then "Users exported"
Show Missions (DataTable)	Click "Show Missions (DataTable)" button	—	Call <code>_loadMissionsForTable(25, 156, example, true)</code> → GET <code>missions_datatable.php?page=1&pageSize=50</code>	Message "Missions: 25 / 156 (example)", DataTable displays
Export Table CSV	Click "Export Table (CSV)" button	non-empty missions	Call <code>_exportMissionsTableCsv()</code> → CSV generation of displayed missions	Download <code>missions_table.csv</code>
Export All CSV	Click "Export all (CSV)" button	—	Call <code>_exportMissionsTableCsv(1256)</code> → GET <code>missions_all.php</code> → full CSV	Message "CSV exported (1256 lines)"
Search missions	Enter in TextField + Enter	—	Update <code>_searchCtrl</code> + reload page 1	DataTable updates

Action	Trigger	Precondition	Effect	Feedback
Change page size	Select dropdown [10, 25, 50, 100, 200]	—	Update <code>_pageSize</code> + reload page 1	DataTable updates
Paginate previous	Click “Prev.”	<code>page > 1</code>	Decrement <code>_page</code> + reload missions	DataTable refreshed
Paginate next	Click “Next”	<code>page < max</code>	Increment <code>_page</code> + reload missions	DataTable refreshed

2.3.4 States & edge cases

- Web only: msg “Export available on web version” if `!kIsWeb`
- Empty DataTable: no table display, “Export Table” button disabled
- Search without results: “No mission” display in DataTable
- Export error → red error message “CSV export error: [error]”
- Loading → buttons disabled, message “Missions: [n] / [total]”
- Page 1: “Prev.” button disabled
- Last page: “Next” button disabled

2.3.5 Important business rules

- CSV generated client-side (Dart) with quote escaping (RFC 4180)
- Full-text search on reference, client, interpreter, product
- Server-side pagination (default 50 per page)
- Download uses `downloadBytes()` (web blob API)
- Interpreter export: all columns of the Interpreter model
- User export: id, username, fullname, email, 4 right flags
- Mission export: 7 key columns (ref, client, interpreter, start, product)

2.3.6 UI dependencies

- `InterpreterService.getInterpreters()`
- `AdminService.getUsers()`
- `MissionService.getMissionsDatatable(), getMissionsDatatableAll()`
- `file_downloader.downloadBytes()`
- `ResponsiveHelper`

- BrandFooter

2.3.7 Notes for the designer (to preserve)

- ElevatedButton.icon buttons with icons (people, person, table_rows, download, download_for_offline)
- Clean screen, no tabs, central actions
- User tip small gray text (color: Colors.black54)
- Fixed 6-column DataTable, horizontal scrollable on mobile

Nom Prénom	Interprètes	Gest. Interprètes	Missions	Admin	Actions
MAJID	☒	☐	☒	☐	✎ ✖
MISBA	☒	☐	☒	☐	✎ ✖
NELLY	☒	☐	☒	☐	✎ ✖
IBRAHIM (De la part de TESFIT)	☒	☐	☒	☐	✎ ✖
Inconnu à renseigner	☒	☒	☒	☐	✎ ✖
Anna ABAD apaula.abad@outlook.fr	☒	☐	☐	☐	✎ ✖
Azhar Abbas azharabbas2k@gmail.com	☒	☐	☐	☐	✎ ✖
Farhat ABBAS abbasfarhat2025@gmail.com	☒	☐	☐	☐	✎ ✖
Natella ABBASSOV n.abbassov@hotmail.fr	☒	☐	☒	☐	✎ ✖
Tarek ABDALJABAR tarekhabdajabar@yahoo.com	☒	☐	☐	☐	✎ ✖
ADEN ABDELKADER	☒	☐	☐	☐	✎ ✖
Asli ABDI	☒	☐	☐	☐	✎ ✖
Adam Abdoulay Abakar	☒	☐	☐	☐	✎ ✖

Figure 2.3: export

2.4 Navigation architecture

2.4.1 Named routes (MaterialApp.routes)

Route	Widget	Access	Precondition
/	HomePage	Tab-based hub	AuthManager.isLogged (default startup)
/login	LoginPage	Login screen	No authentication required (fallback to home if not logged)
/admin-mockup	AdminMockupPreview	Mockup only (debug/design)	No condition

Route	Widget	Access	Precondition
/interpreters	InterpretersPage	Interpreter management	canManageInterpreters() \ \ isAdmin()
/missions-table	MissionsTablePage	Mission tracking	canManageMissions() \ \ isAdmin()
/billing	BillingPage	Billing	canManageMissions() \ \ isAdmin()
/admin	AdminPage	User admin	isAdmin()
/export	ExportPage	Data export	isAdmin()
/company-info	CompanyInfoPage	Company info	canManageInterpreters() \ \ isAdmin()
/requesters	RequestersManagementPage	Third parties management	canManageMissions() \ \ isAdmin()

2.4.2 Roles × Accessible pages matrix

Role	/interpreters	/missions-table	/billing	/requesters	/admin	/export	/company-info
No rights	☐	☐	☐	☐	☐	☐	☐
interpreter	☐	☐	☐	☐	☐	☐	☐
agent_admin_annuaire (In-ter-preter Mgr.)	☐	☐	☐	☐	☐	☐	☐
agent_admin_mission (Mis-sion Mgr.)	☐	☐	☐	☐	☐	☐	☐
admin	☐	☐	☐	☐	☐	☐	☐
admin + interpretate	☐	☐	☐	☐	☐	☐	☐

2.4.3 Authentication flow

1. **App startup** → [lib/main.dart](#) calls `AuthManager.restoreSession()`
2. **Valid session** → `MyApp(startOnHome: true)` → `HomePage` route (default `TabBar`)
3. **No session** → `MyApp(startOnHome: false)` → `LoginPage` route
4. **Login submit** → `AuthService.login()` POST `/api/login.php`
5. **Success** → `AuthManager.setUser() + setRights() + persistSession()` → `HomePage`
6. **Logout** → `AuthManager.logout() + pushReplacementNamed('/login')`
7. **Remember me** → session persisted via `SharedPreferences` (re-auto-login next session)

2.4.4 Runtime access control (UserRights)

- **isAdmin()**: contains 'admin' OR 'agent_admin' → access to `/admin`, `/export`, all modules

- **canManageInterpreters()**: isAdmin() OR contains 'agent_admin_annuaire' → access to /interpreters, /company-info
- **canManageMissions()**: isAdmin() OR contains 'agent_admin_mission' → access to /missions-table, /billing, /requesters
- **isInterpreter()**: contains 'interprete' → conditional interpreter UI display (no access to separate pages)

2.4.5 Modal dialogs & confirmations

Screen	Dialog	Opening	Effect
AdminPage	User edit	Click Edit	Modify name, email, login, password + POST update_user.php
AdminPage	Deletion confirmation	Click Delete	Confirmation then POST delete_user.php + list reload
HomePage	Logout confirmation	Click Logout button	AuthManager.logout() + return to /login

2.4.6 Global state (AuthManager) — singleton scope

- `_user`: active user Map (id, prenom, nom, login)
- `_rights`: UserRights instance (rights list)
- `_remember`: boolean session persistence
- SharedPreferences keys: `auth_user`, `auth_rights`, `auth_remember`

Chapter 3

2. Directory (companies, contacts, interpreters)

3.1 Add/Edit a Company — [lib/pages/requesters_management_page.dart](#)

Route / access: From Companies Management page → « Add » or « Edit » button

Business purpose: Create/edit form for a requester company with its identity, contact and location data.

API endpoints: [api/add_client.php](#), [api/update_client.php](#), [api/get_countries.php](#), [api/get_departments.php](#)

3.1.1 Layout & zones

- **Title:** « New requester company » or « Edit company »
- **Content:** SingleChildScrollView containing Column of TextField/DropdownButtonFormField
- **Actions:** « Cancel » and « Save » buttons
- **Errors:** Red text display below the form

3.1.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Name *	TextInput	String	—	Required
Alias	TextInput	String	—	Optional
Address	TextInput	String	—	Optional
Postal code	TextInput (Row 50%)	String	—	Optional
City	TextInput (Row 50%)	String	—	Optional
Country	Dropdown	Country ID	—	Optional, with matching by label/code
Department	Dropdown	Department ID	—	Optional, with matching by label/code
Phone	TextInput	String	—	Optional
Fax	TextInput	String	—	Optional
Email	TextInput	String	—	Optional
Website	TextInput	String	—	Optional
SIREN	TextInput (Row 50%)	String	—	Optional

Field	Source	Format	Sort/Filter	Notes
SIRET	TextInput (Row 50%)	String	—	Optional
Public note	TextArea	String (2-4 lines)	—	Optional
Private note	TextArea	String (2-4 lines)	—	Optional

3.1.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Save	Click « Save » button	Non-empty Name	POST/PUT to API → List reload	Success Snackbar
Cancel	Click « Cancel » button	—	Closes dialog without saving	Return to previous page
Load countries	Init dialog	—	Call <code>CountryService.getCountryes()</code>	Dropdown populated
Load departments	Init dialog	—	Call <code>DepartmentService.getDepartments()</code>	Dropdown populated
Change country	Country dropdown selection	—	Updates local <code>countryId</code>	Stateful control
Change department	Department dropdown selection	—	Updates local <code>departmentId</code>	Stateful control

3.1.4 States & edge cases

- **Creation mode:** All fields empty except country/department if detected
- **Edit mode:** Pre-fill all fields with current values
- **Name validation:** Error « Company name is required. » if empty
- **API error:** Red error message display, `saving=false` state restored
- **Country not found:** Displays « Not specified » in dropdown, plain label fallback
- **Department not found:** Displays « Not specified » in dropdown, plain label fallback

3.1.5 Important business rules

1. **Country/department matching:** Performed at build via `_matchCountryId()` and `_matchDepartmentId()` which test existing ID, label, code, code_iso
2. **ID auto-assigned:** New company gets auto-generated ID from server
3. **User tracking:** `user_id` field populated via `AuthManager.userId` if > 0
4. **Creation status:** New companies created with `status=1` (active) by default
5. **Update mode:** Preserves the current status of the company (can be disabled separately)

3.1.6 UI dependencies

- `lib/models/country.dart`: Country
- `lib/models/department.dart`: Department
- `lib/services/country_service.dart`: `CountryService.getCountryes()`
- `lib/services/department_service.dart`: `DepartmentService.getDepartments()`
- `lib/services/client_service.dart`: `ClientService`

3.1.7 Notes for the designer (to preserve)

- **Width:** AlertDialog width=560px

- **Spacing:** SizedBox(height: 10) between fields
- **Rows:** Some fields grouped 50/50 with SizedBox(width: 12) between
- **Validation:** « Save » button disabled while saving=true (MonoColor progress)
- **Labels:** « * » for required on the « Name » field only
- **Hint texts:** No hint, just labels

The screenshot shows the 'Sociétés et contacts' page. On the left, there's a list of companies under 'Sociétés demandeuses'. The selected company is '1270 NOTAIRES - 0732'. On the right, the details for this company are shown, including fields for Alias, Ville, Département, Fax, Site web, SIRET, Adresse, Note publique, Note privée, Code postal, Pays, Téléphone, Email, and SIREN. There are buttons for 'Modifier' and 'Désactiver'.

Figure 3.1: requesters-add-edit-company

3.2 Confirm Company Deletion — [lib/pages/requesters_management_page.dart](#)

Route / access: From Company Details → « Deactivate » button

Business purpose: Confirm deletion (soft delete) of a company.

API endpoints: [api/delete_client.php](#)

3.2.1 Layout & zones

- **Title:** « Delete company »
- **Content:** Text « The company "[name]" will be deactivated in Dolibarr. Continue? »
- **Actions:** « Cancel » and « Deactivate » buttons

3.2.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Company name	company.name	Text (dynamic)	—	Embedded in the message

3.2.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Confirm deactivation	Click « Deactivate »	—	Call DELETE API → List reload	SnackBar « Company deactivated. »
Cancel	Click « Cancel »	—	Closes dialog without action	Return to previous page

3.2.4 States & edge cases

- **API error:** Displays error SnackBar, no dialog closing
- **Lost mounting:** Checks `!mounted` before UI update

3.2.5 Important business rules

1. **Soft delete:** API call sets `status = 0` (no physical deletion)
2. **User tracking:** `user_id` field populated via `AuthManager.userId` if `> 0`

3.2.6 UI dependencies

- [lib/services/client_service.dart](#): `ClientService.deleteRequestingCompany()`

3.2.7 Notes for the designer (to preserve)

- **Dialog:** Standard `AlertDialog`
- **Danger button:** « Deactivate » = red `FilledButton` (color `#E1000F`)

3.3 Add/Edit a Contact — [lib/pages/requesters_management_page.dart](#)

Route / access: From Contacts Panel → « Add » button or « Edit » icon

Business purpose: Create/edit form for a requester person (contact) attached to a company.

API endpoints: [api/add_contact.php](#), [api/update_contact.php](#), [api/get_countries.php](#), [api/get_departments.php](#)

3.3.1 Layout & zones

- **Title:** « New requester person » or « Edit requester person »
- **Content:** `SingleChildScrollView` containing `Column` of `TextField/DropdownButtonFormField`
- **Actions:** « Cancel » and « Save » buttons
- **Errors:** Red text display below the form

3.3.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Civility	<code>TextInput</code> (Row 33%)	String	—	Optional, e.g. « Mr », « Mrs »
First name	<code>TextInput</code> (Row 33%)	String	—	Optional
Last name	<code>TextInput</code> (Row 33%)	String	—	Optional
Function	<code>TextInput</code>	String	—	Optional
Email	<code>TextInput</code>	String	—	Optional

Field	Source	Format	Sort/Filter	Notes
Phone	TextInput (Row 50%)	String	—	Optional
Personal phone	TextInput (Row 50%)	String	—	Optional
Mobile	TextInput (Row 50%)	String	—	Optional
Fax	TextInput (Row 50%)	String	—	Optional
Date of birth	TextInput	String (YYYY-MM-DD format)	—	Optional
Address	TextInput	String	—	Optional
Postal code	TextInput (Row 50%)	String	—	Optional
City	TextInput (Row 50%)	String	—	Optional
Country	Dropdown	Country ID	—	Optional
Department	Dropdown	Department ID	—	Optional
Public note	TextArea	String (2-4 lines)	—	Optional
Private note	TextArea	String (2-4 lines)	—	Optional

3.3.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Save	Click « Save » button	Non-empty First name OR Last name	POST/PUT to API → Contacts reload	Success Snackbar
Cancel	Click « Cancel » button	—	Closes dialog without saving	Return to previous page
Load countries	Init dialog	—	Call <code>CountryService.getCountryCodes()</code>	Dropdown populated
Load departments	Init dialog	—	Call <code>DepartmentService.getDepartments()</code>	Dropdown populated

3.3.4 States & edge cases

- **Creation mode:** All fields empty except country/department if detected
- **Edit mode:** Pre-fill all fields with current values
- **Name validation:** Error « First name or last name is required. » if both empty
- **API error:** Red error message display, `saving=false` state restored
- **Country not found:** Displays « Not specified » in dropdown, plain label fallback
- **Department not found:** Displays « Not specified » in dropdown, plain label fallback

3.3.5 Important business rules

1. **Validation:** At least one of first name/last name required (validated by custom FormField)
2. **Country/department matching:** Performed at build via the same functions as the company
3. **Department display:** Format « [CODE] - [LABEL] » if code present
4. **Parent company:** Contact always linked to `_selectedCompanyId` (precondition)
5. **User tracking:** `user_id` field populated via `AuthManager.userId` if `> 0`
6. **Creation status:** New contacts created with `status=1` (active)

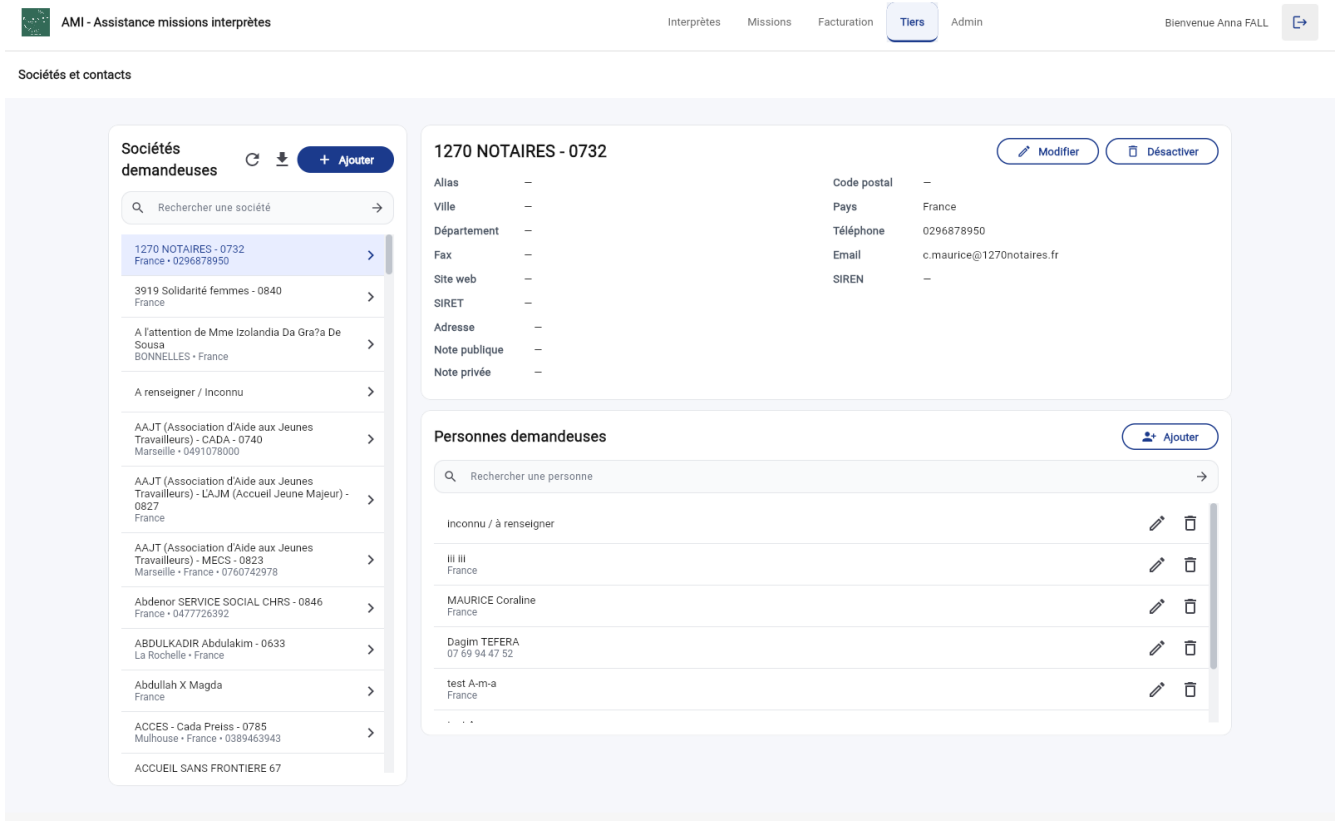


Figure 3.2: requesters-delete-company-confirm

3.3.6 UI dependencies

- [lib/models/country.dart](#): Country
- [lib/models/department.dart](#): Department
- [lib/services/contact_service.dart](#): ContactService

3.3.7 Notes for the designer (to preserve)

- **Width:** AlertDialog width=560px
- **Spacing:** SizedBox(height: 10) between fields
- **Rows:** Some fields grouped 50/50 or 33/33/33 with SizedBox(width: 12) between
- **Validation:** « Save » button disabled while saving=true
- **Date hint:** « YYYY-MM-DD »
- **Civility hint:** « Mr, Mrs, Dr... »

3.4 Confirm Contact Deletion — [lib/pages/requesters_management_page.dart](#)

Route / access: From Contacts Panel → « Delete » icon

Business purpose: Confirm deletion (soft delete) of a contact.

API endpoints: [api/delete_contact.php](#)

3.4.1 Layout & zones

- **Title:** « Delete requester person »

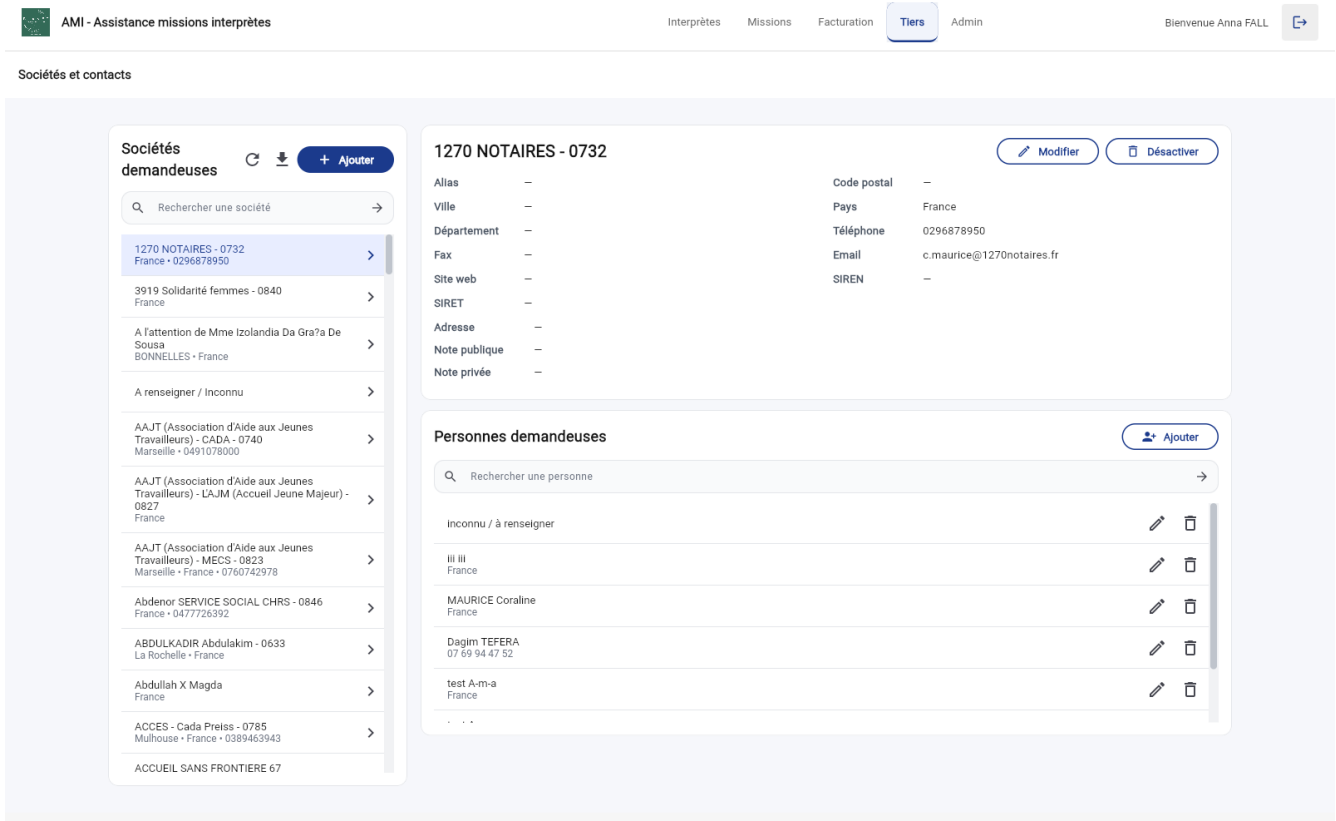


Figure 3.3: requesters-add-edit-contact

- **Content:** Text « The contact "[displayName]" will be deactivated in Dolibarr. Continue? »
- **Actions:** « Cancel » and « Deactivate » buttons

3.4.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Contact name	<code>contact.displayNameText</code> (dynamic)	—	—	Embedded in the message

3.4.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Confirm deactivation	Click « Deactivate »	—	Call DELETE API → Contacts reload	SnackBar « Contact deactivated. »
Cancel	Click « Cancel »	—	Closes dialog without action	Return to previous page

3.4.4 States & edge cases

- **API error:** Displays error SnackBar, no dialog closing
- **Lost mounting:** Checks `!mounted` before UI update

3.4.5 Important business rules

1. **Soft delete:** API call sets status = 0 (no physical deletion)
2. **User tracking:** user_id field populated via AuthManager.userId if > 0

3.4.6 UI dependencies

- [lib/services/contact_service.dart](#): `ContactService.deleteContact()`

3.4.7 Notes for the designer (to preserve)

- **Dialog:** Standard AlertDialog
- **Danger button:** « Deactivate » = red FilledButton (color #E1000F)

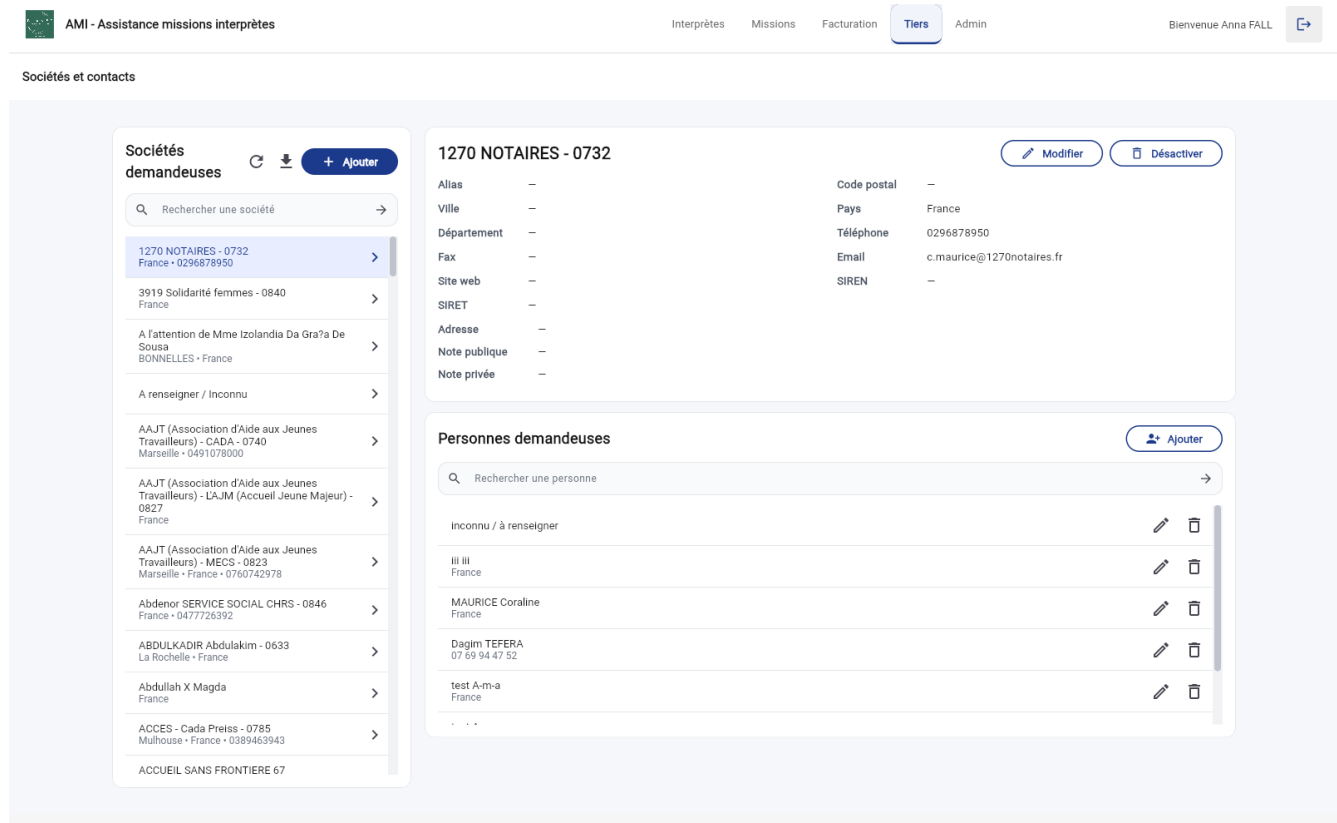


Figure 3.4: requesters-delete-contact-confirm

3.5 Interpreter Management — [lib/pages/interpreters_page.dart](#)

Route / access: Main menu → Interpreter management

Business purpose: Manage the interpreter directory with search, add, edit, delete and contact actions (call, WhatsApp). Responsive grid display.

API endpoints: [api/get_interpretes.php](#), [api/add_interprete.php](#), [api/update_interprete.php](#), [api/delete_interprete.php](#)

3.5.1 Layout & zones

- **Search/actions bar:** Fixed container at the top with gray background #F8FAFC

- Search field (focus border #000091)
- Buttons: FMI (TextButton/IconButton depending on size), CSV (TextButton/IconButton), Add (Elevated-Button/IconButton)
- **Main area:** Responsive GridView or ListView depending on screen size
 - Mobile (< 600px): 1 column, aspect ratio 0.85
 - Portrait tablet (600–900px): 2 columns, aspect ratio 0.95
 - Landscape tablet (900–1400px): 3 columns, aspect ratio 1.0
 - Desktop (1400–1800px): 4 columns, aspect ratio 1.1
 - Large desktop (1800–2200px): 5 columns, aspect ratio 1.2
 - Extra large desktop (>= 2200px): 6 columns, aspect ratio 1.3
- **Interpreter cards:** Padding 12–16px, border #E5E7EB, borderradius 4px
- **Footer:** Standard BrandFooter

3.5.2 Data displayed (columns / fields)

3.5.2.1 Search bar

Field	Source	Format	Sort/Filter	Notes
Search	Input text	String	Local filter on name, languages, city	Placeholder « Search an interpreter »

3.5.2.2 Interpreter card

Field	Source	Format	Sort/Filter	Notes
Name (title)	Interpreter.displayname	Text 20pt bold uppercase	—	Centered
Languages	Interpreter.languages	Full case text	—	Centered, optional « Not specified »
Phone	Priority mobile > home	Blue #000091 clickable TextButton	—	Format « XX XX XX XX », centered
Email	Interpreter.email	Gray #666666 text with icon	—	Centered, ellipsis overflow
Status	Interpreter.status	Badge with color	—	Available = green #B8FEC9, On mission = yellow #FFF0CC, Unavailable = red #FFE9E9
Notes	Interpreter.comments	Small text (optional)	—	« See more » button if present
Actions	Buttons	Colored icons	—	WhatsApp green #25D366, Edit blue #000091, Delete red #E1000F

3.5.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Search	Type in field or click reset icon	—	Local filter on interpreter list	Real-time filtered list display
Open FMI	Click « FMI » button	—	Opens URL <code>https://fmi.planetapplis.fr/</code> in external app	SnackBar if error
Export CSV	Click « CSV » button	Loaded interpreters	Generates <code>interpretes.csv</code> file	Success/error SnackBar
Add interpreter	Click « Add » button	<code>canManageInterpreters()</code> or <code>isAdmin()</code> rights	Opens Dialog 1: Add Interpreter	AlertDialog
Edit interpreter	Click « Edit » button on card	<code>canManageInterpreters()</code> rights	Opens Dialog 1: Edit Interpreter	DialogBox with pre-fill
Delete interpreter	Click « Delete » button on card	<code>canManageInterpreters()</code> rights	Opens Dialog 2: Confirm Deletion	Confirmation DialogBox
Call interpreter	Click on phone number	Number present	Opens system call app with <code>tel: URI</code>	SnackBar if error
Send WhatsApp	Click green WhatsApp button	Number present	Opens WhatsApp web <code>https://wa.me/[number]</code>	SnackBar if error
See details	Click « See more » button	Notes present	Opens Dialog 3: Interpreter Details	AlertDialog with notes
Keyboard scroll	Arrow Up/Down, Page Up/Down, Home, End	Focus on page	Continuous animated scroll	Animated with easing curves

3.5.4 States & edge cases

- **No interpreter found:** Displays « No interpreter found » in the center
- **Initial load:** `CircularProgressIndicator`
- **Empty filter:** Displays all interpreters
- **No number:** « Not specified » in gray italic, call/WhatsApp buttons disabled
- **No email:** « Not specified » in gray italic
- **No notes:** « See more » button hidden
- **Load error:** Displays « Error: [message] »

3.5.5 Important business rules

1. **Local filter:** Search on `displayName`, `languesParlees`, `ville` (case-insensitive)
2. **Interpreter statuses:** « Available », « Unavailable », « On mission » (normalized via `normalizeStatus()`)
3. **Phone format:** Display « XX XX XX XX » (2-digit groups); cleaning before call/WhatsApp (regex `[^0-9+]`)
4. **WhatsApp:** GET request to `https://wa.me/[number_without_symbols]`; + prefix or direct
5. **Display rights:** Edit/Delete buttons hidden if no rights, visible if `canManageInterpreters()` or `isAdmin()`
6. **CSV export:** Includes: `numero`, `nom`, `prenom`, `langues`, `email`, `tel_mobile`, `tel_domicile`, `adresse`, `code_postal`, `ville`, `pays`, `statut`, `commentaires`

3.5.6 UI dependencies

- `lib/models/interpreter.dart`: `Interpreter`
- `lib/models/country.dart`: `Country`
- `lib/services/interpreter_service.dart`: `InterpreterService`
- `lib/services/country_service.dart`: `CountryService`

- [lib/core/user_rights.dart](#): UserRights
- [lib/core/responsive_helper.dart](#): responsive helpers
- [lib/core/csv_export.dart](#): downloadCsv()
- [lib/core/brand_footer.dart](#): BrandFooter
- External: [url_launcher](#) for calls/WhatsApp/URLs

3.5.7 Notes for the designer (to preserve)

- **Keyboard scroll**: Page Down/Up scroll by 80% height, Space = Page Down, Home/End scroll to extremes
- **Keyboard focus**: FocusNode on build, focus request after first frame
- **Grid spacing**: Variable horizontal padding depending on screen size (16–64px)
- **Status badge**: 1px border + solid background
- **Action icons**: 20px size, white foreground on colored backgrounds
- **Breakpoints**: 576px (mobile), 768px (tablet), 1200px (desktop), 1440px (large), 1800px (extra large)

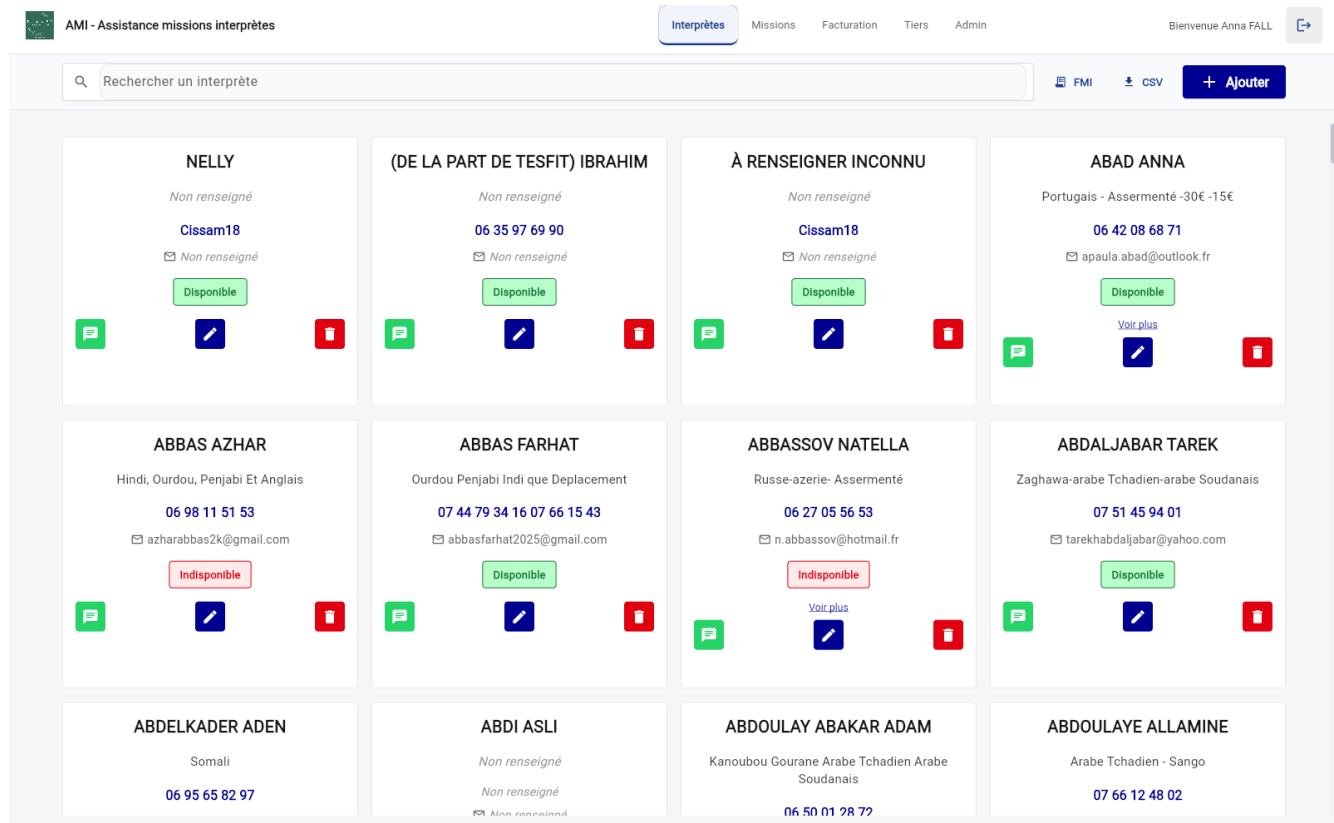


Figure 3.5: interpreters-main

3.6 Add/Edit an Interpreter — [lib/pages/interpreters_page.dart](#)

Route / access: From Interpreters page → « Add » button or « Edit » button on card

Business purpose: Create/edit form for an interpreter with identity, contact and status data.

API endpoints: [api/add_interprete.php](#), [api/update_interprete.php](#), [api/get_countries.php](#)

3.6.1 Layout & zones

- **Title**: « Add an interpreter » or « Edit an interpreter »

- **Content:** SingleChildScrollView containing Form with TextFormField/DropdownButtonFormField
- **Actions:** « Cancel » and « Save » buttons
- **Validation:** Errors displayed in red below the form (custom FormField)

3.6.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Last name	TextFormField	String	—	Required (at least last name OR first name)
First name	TextFormField	String	—	Optional
Validation of at least one	Custom FormField	Error message	—	Displays « Provide at least last name or first name »
Mobile phone	TextFormField	String	—	Optional
Landline phone	TextFormField (label « Landline »)	String	—	Optional
Email	TextFormField	String (keyboard-Type.emailAddress)	—	Optional
Spoken languages	TextFormField	Multi-line String (2 lines)	—	Optional
Status	DropdownButtonFormField	Selection among « Available », « Unavailable », « On mission »	—	Required, default « Available »
Country	DropdownButtonFormField	Country ID	—	Optional
Notes	TextFormField	Multi-line String (3 lines)	—	Optional

3.6.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Load countries	Init dialog	—	Call <code>CountryService.getPopulated()</code>	Dropdown populated
Edit field	Type in TextFormField	—	Local validation if <code>!isEdit</code>	Call <code>formKey.currentState?.validate()</code>
Change status	Dropdown selection	—	Updates local <code>statusValue</code>	Stateful control
Change country	Dropdown selection	—	Updates local <code>countryId</code>	Stateful control
Save	Click « Save »	Form validation OK	POST/PUT to API → Reload list	SnackBar « Interpreter created/updated »
Cancel	Click « Cancel »	—	Closes dialog without action	Return to previous page

3.6.4 States & edge cases

- **Creation mode:** All fields empty except default status « Available »
- **Edit mode:** Pre-fill with current values
- **Validation:** Error « Provide at least last name or first name » if both empty (custom FormField)
- **API error:** Error message display + `saving=false` state restored

- **Unrecognized country:** Label used in plain text fallback
- **Login generation:** Performed server-side (add_interprete.php) via slugifyLogin(lastname+getFirstLetter(firstname))
- **Password generation:** Server-side, format [lastname]2026

3.6.5 Important business rules

1. **Validation:** At least one of last name/first name required (custom FormField)
2. **Name/first name resolution:** Complex fallback logic for display pre-fill (resolvedNom, resolvedPrenom)
3. **displayName:** Final uppercase format « LASTNAME FIRSTNAME »
4. **Country matching:** Via fkCountry or label/code; fallback on plain label
5. **Login creation:** Concatenation lastname + first letter firstname, slugified, unique via numeric suffix
6. **Password creation:** Bcrypt hash server-side, plaintext returned for initial transmission
7. **Interpreter right:** Automatic assignment via ensureInterpreterRight() on creation
8. **User tracking:** No user_id sent (interpreter = created user, not admin action)

3.6.6 UI dependencies

- [lib/models/interpreter.dart](#): Interpreter
- [lib/models/country.dart](#): Country
- [lib/services/interpreter_service.dart](#): InterpreterService
- [lib/services/country_service.dart](#): CountryService
- [lib/core/responsive_helper.dart](#): ResponsiveHelper.getDialogWidth()

3.6.7 Notes for the designer (to preserve)

- **Dialog width:** ResponsiveHelper.getDialogWidth(context)
- **Max height:** 85% of screen height
- **Spacing:** By default FlutterTextField isDense=true
- **Validation display:** Custom FormField for error message above last name/first name fields
- **Dropdown items:** « Not specified » null-option for country
- **Status options:** « Available », « Unavailable », « On mission »

3.7 Confirm Interpreter Deletion — [lib/pages/interpreters_page.dart](#)

Route / access: From Interpreter card → « Delete » button

Business purpose: Confirm deletion (soft delete) of an interpreter.

API endpoints: [api/delete_interprete.php](#)

3.7.1 Layout & zones

- **Title:** « Delete »
- **Content:** Text « Delete [displayName]? »
- **Actions:** « Cancel » and « Delete » buttons

3.7.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Interpreter name	Interpreter.displayName	Text (dynamic)	—	Embedded in the message

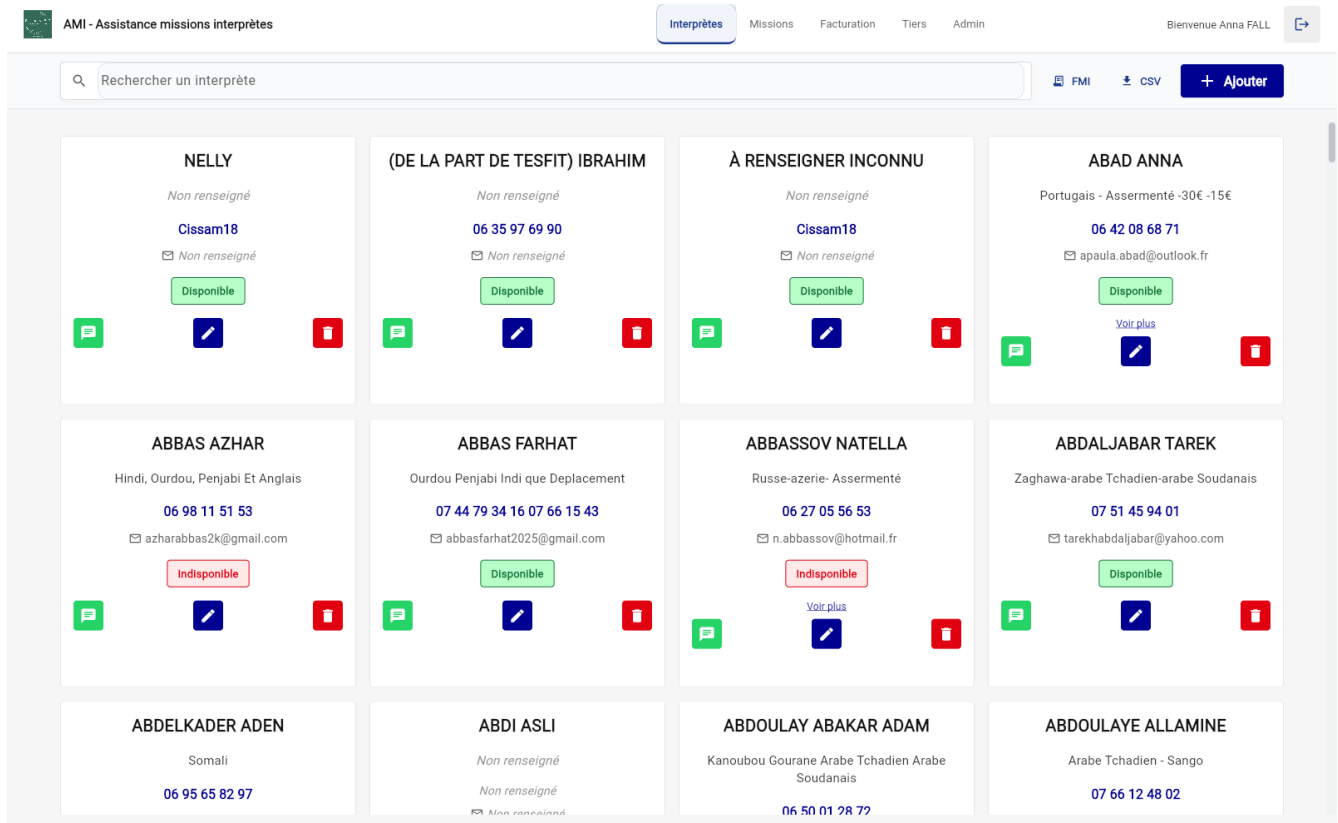


Figure 3.6: interpreters-add-edit-interpreter

3.7.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Confirm deletion	Click « Delete »	—	Call DELETE API → List reload	Success or error SnackBar
Cancel	Click « Cancel »	—	Closes dialog without action	Return to previous page

3.7.4 States & edge cases

- **API error:** Displays error SnackBar
- **Lost mounting:** Checks !mounted before update

3.7.5 Important business rules

1. **Soft delete:** API call sets statut = -1 (no physical deletion)
2. **FK preservation:** Soft delete to avoid FK constraint errors from invoices, missions, etc.

3.7.6 UI dependencies

- `lib/services/interpreter_service.dart`: `InterpreterService.deleteInterpreter()`

3.7.7 Notes for the designer (to preserve)

- **Dialog:** Dialog wrapping AlertDialog

- **Dialog dimensions:** Responsive width + max height 70% viewport
- **Danger button:** « Delete » = red ElevatedButton #E1000F

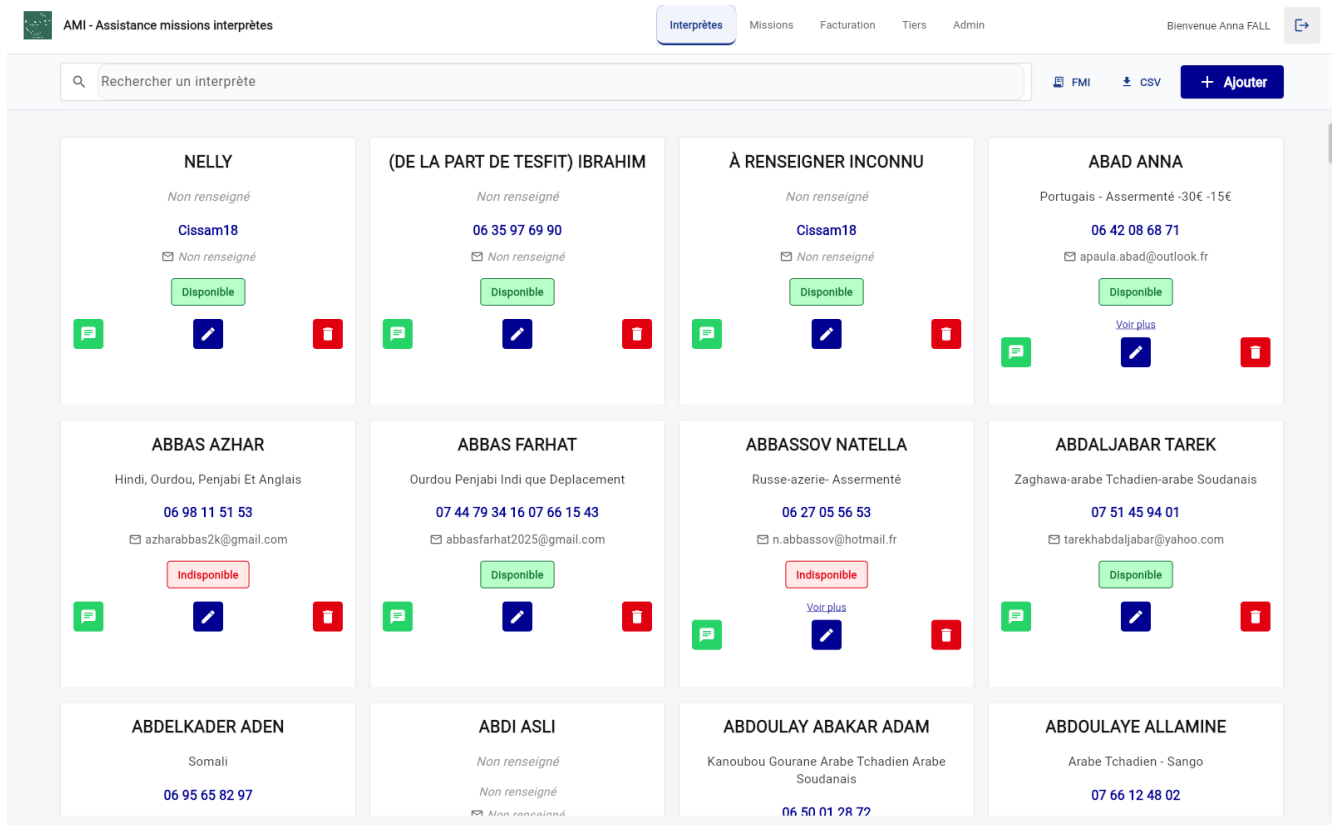


Figure 3.7: interpreters-delete-interpreter-confirm

3.8 Interpreter Details — [lib/pages/interpreters_page.dart](#)

Route / access: From Interpreter card → « See more » button (if notes present)

Business purpose: Display complete details of an interpreter, notably languages and notes.

API endpoints: None (local data)

3.8.1 Layout & zones

- **Title:** `Interpreter.displayName` centered, bold 18pt
- **Content:** `SingleChildScrollView` containing `Column`
 - Languages (title-case, centered if present)
 - Divider
 - Label « Notes: » gray 13pt
 - Notes text
- **Actions:** « Close » button

3.8.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Name (title)	<code>Interpreter.displayName</code>	Text 18pt bold	—	Centered

Field	Source	Format	Sort/Filter	Notes
Languages	Interpreter.languages	Title case text	—	Centered, if present
Notes	Interpreter.comments	Plain text 14pt	—	After divider

3.8.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Close dialog	Click « Close »	—	Closes AlertDialog	Return to previous page

3.8.4 States & edge cases

- **No language:** Does not display the languages section
- **No notes:** Displays dialog anyway (rarely, since dialog only opens if notes)

3.8.5 Important business rules

1. **Conditional display:** Languages displayed only if non-empty
2. **Text normalization:** `_toTitleCase()` for languages

3.8.6 UI dependencies

- [lib/models/interpreter.dart](#): Interpreter

3.8.7 Notes for the designer (to preserve)

- **Dialog:** Standard AlertDialog
- **Spacing:** `SizedBox(height: 12)` between sections
- **Divider:** Standard MaterialUI
- **Text styling:** Gray #3A3A3A color for languages/notes

3.9 Interpreter Missions — [lib/pages/interpreter_missions_page.dart](#)

Route / access: From Interpreter card or listing → « See missions » action

Business purpose: List and filter the missions of a specific interpreter by year/month. Table vs list depending on screen size.

API endpoints: [api/get_missions_by_interpreter.php](#) (via `MissionService.getMissionsByInterpreter()`)

3.9.1 Layout & zones

- **AppBar:** « Missions of [interpreterName] »
- **Filters:** Centered Row with year/month dropdowns and « Reset » button
- **Main area:** ListView or DataTable depending on screen size
 - Mobile (< 800px): ListView of cards
 - Desktop (>= 800px): Scrollable DataTable (H + V)
- **Table columns:** Reference, Product, Start, End, Payment

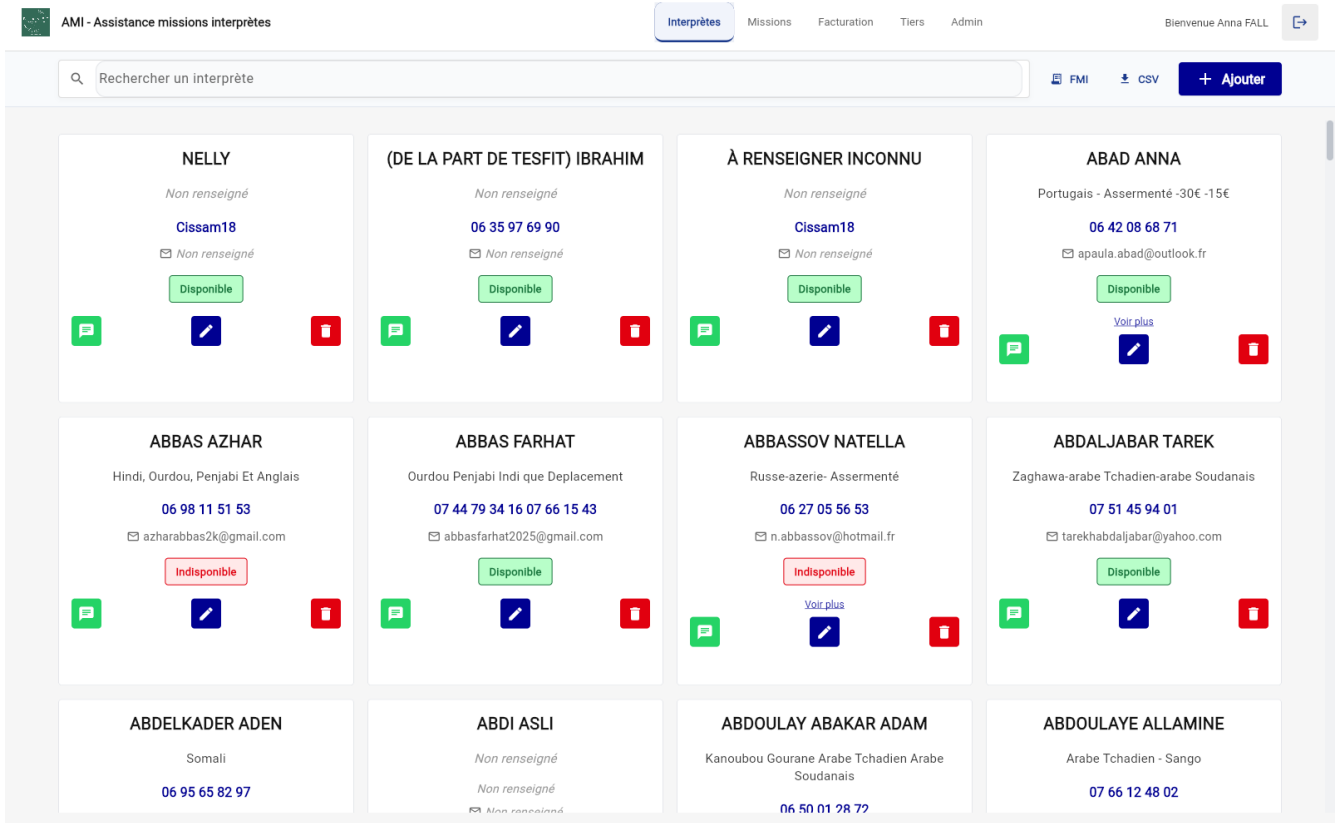


Figure 3.8: interpreters-details-dialog

3.9.2 Data displayed (columns / fields)

3.9.2.1 Filters

Field	Source	Format	Sort/Filter	Notes
Year	Dropdown	List unique years	Filter	« All » = null
Month	Dropdown	List months 1–12	Filter	« All » = null, format « January », « February », etc. (locale fr_FR)

3.9.2.2 Mission Row/Card (mobile)

Field	Source	Format	Sort/Filter	Notes
Reference	mission['reference_dev']	Clickable text (copy icon)	—	If non-empty and != « No reference »/« N/A »
Product	mission['ref']	Text	—	—
Start	mission['debutmission']	Date format dd/MM/yyyy	—	—
End	mission['finmission']	Date format dd/MM/yyyy	—	—
Payment	mission['billed_status']	Gray chip	—	Billing status

3.9.2.3 DataTable (desktop)

Column	Source	Format	Sort/Filter	Notes
Reference	mission['reference_dev']	Text + copy icon	—	Copyable on click
Product	mission['ref']	Text	—	—
Start	mission['debutmission']	Date format dd/MM/yyyy	—	—
End	mission['finmission']	Date format dd/MM/yyyy	—	—
Payment	mission['billed_status']	Text	—	—

3.9.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Load missions	Init page	—	Call GET missions API	FutureBuilder displays content
Filter by year	Year dropdown selection	—	Updates _filterYear → local filter	Displays missions for the year
Filter by month	Month dropdown selection	—	Updates _filterMonth → local filter	Displays missions for the month
Reset filters	Click « Reset » button	—	Resets _filterYear and _filterMonth to null	Displays all missions
Copy reference	Click copy icon on reference	Non-empty reference	Copy to clipboard	SnackBar « Reference copied »

3.9.4 States & edge cases

- **No mission:** « No mission found for this interpreter. »
- **No mission after filter:** Displays empty list
- **Load error:** « Error: [message] »
- **Invalid reference:** Copy icon hidden if empty, « No reference », or « N/A »

3.9.5 Important business rules

1. **Local filter:** Performed in memory after loading (no re-query API)
2. **Derived years:** Extracted from all missions via `_parseDate(mission['debutmission']).year`
3. **Date format:** Uses `DateFormat('dd/MM/yyyy')` from `intl` package
4. **Month locale:** « fr_FR » for displaying month names in French
5. **Responsive breakpoint:** 800px = ListView → DataTable switch
6. **Reference copy:** Via `Clipboard.setData()`

3.9.6 UI dependencies

- `lib/services/mission_service.dart`: `MissionService.getMissionsByInterpreter()`
- `intl` package: `DateFormat`
- `flutter/services`: `Clipboard`

3.9.7 Notes for the designer (to preserve)

- **Filter row:** Centered, Theme VisualDensity.compact
- **DataTable:** Bidirectionally scrollable if wide columns
- **Spacing:** SizedBox(width: 8) between dropdowns and button
- **Payment chip:** blueGrey.shade100 gray background, dark gray text
- **Copy icon:** 18px size, blue #000091, tooltip « Copy »
- **Mobile card:** Standard padding, ListTile for title/subtitle, isThreeLine=true

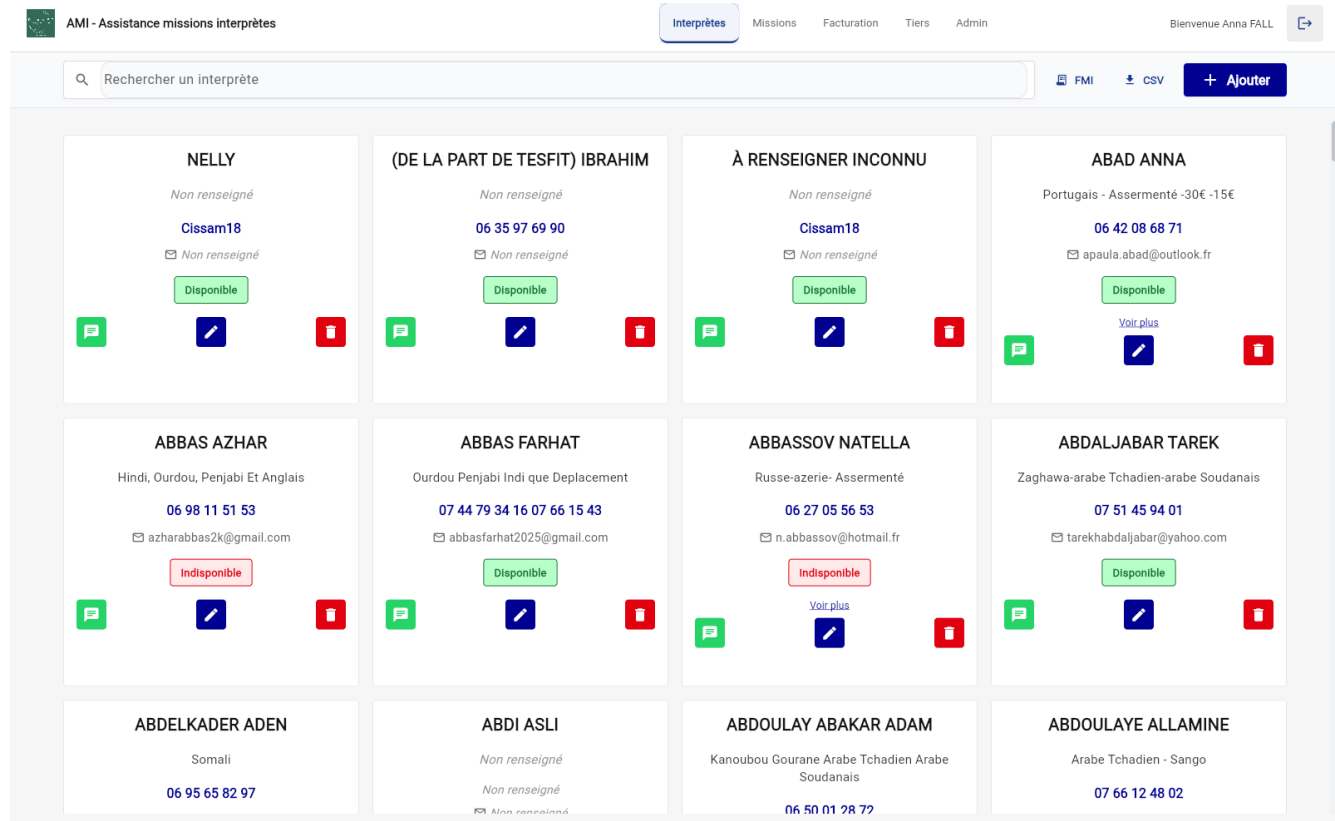


Figure 3.9: interpreters-missions

Chapter 4

3. Missions

4.1 Mission Form Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Modal dialog opened by clicking edit button (table actions) or via embedded form in “New mission” sidebar

Business purpose: Complete form to create or edit a mission. Captures all business fields: interpreter, client/contact, date, time, duration, language, mission type, status, label, notes. Displays a summary created after success.

API endpoints: - POST `api/add_mission_interpreter.php` — mission creation (payload: interpreter_id, client_id, contact_id, label, date, mission, heuredebutmission, dureemission, produit_ref, mission_types, mission_status, commentaires, debutmission, finmission, creator_id) - POST `api/update_mission_interpreter.php` — mission update (id + fields to update) - POST `api/get_interpretes.php` (via MissionService) — interpreter lookup with search - POST `api/get_clients.php` (via ClientService) — client lookup with search - GET/POST `api/get_contacts.php` (via ContactService) — contacts per client - GET `api/get_languages.php` (via LanguageService) — language lookup with search

4.1.1 Layout & zones

- **Dialog wrapper:** `Dialog(insetPadding: 24px, maxWidth: 900px, maxHeight: 720px)`
- **Embedded variant:** Card in Container (“New mission” sidebar)
- **Content:** `Scrollbar + SingleChildScrollView` (vertical scroll if form > 720px)
- **Visible sections:**
 - Loaders: `LinearProgressIndicator` during async lookup
 - Validation error: Red Container at top if message
 - Success summary: Displays snapshot of created mission (if `_showCreatedSummary=true` and embedded)
 - “Mission type” block: Multi-choice `FilterChip` (Judicial Court, Translation, Interpretation)
 - “Label” block: Simple `TextField`
 - “Planning” block: Date (date picker), Time (time picker), Duration (text input HH:mm or minutes)
 - “Actors” block: 3 pickers (Company, Contact, Interpreter)
 - “Mission details” block: Language (dropdown/picker), Status (dropdown/picker), Notes (multi-line)
 - Actions: Cancel button, Submit (disabled if required fields empty or `submitting=true`)

4.1.2 Data displayed (columns / fields)

Field	Source	Format	Sort/Filter	Notes
Mission type	<code>_selectedMissionType</code> Set	Multi-select <code>FilterChip</code>	Filter N/A; Sort N/A	Default “Interpretation” if empty

Field	Source	Format	Sort/Filter	Notes
Label	_labelCtrl TextEditingController	Text (max ?)	Filter N/A; Sort N/A	Optional
Mission date	_selectedMissionDate DateTime	Date picker UI	Filter N/A; Sort N/A	Display format dd/MM/yyyy
Start time	_selectedMissionTime TimeOfDay	Time picker UI	Filter N/A; Sort N/A	Display format HH:mm
Duration	_dureeCtrl + _parseDurationMinutes	Minutes (input: "30", "2h30", "2:30", "150")	Filter N/A; Sort N/A	Derived start→end if provided
Requester company	_selectedClientId int + _clientCtrl text	Autocomplete picker dialog	Filter N/A (lookup-only); Sort N/A	ClientService lookup
Requester person	_selectedContactId int + _contactCtrl text	Picker dialog (client contacts)	Filter N/A; Sort N/A	ContactService lookup (loaded after client)
Interpreter	_selectedInterpreter int + _interpreterCtrl text	Picker dialog	Filter N/A; Sort N/A	MissionService lookup
Language	_selectedLanguageRef string + _langueCtrl text	Picker dialog	Filter N/A; Sort N/A	LanguageService lookup (product ref)
Mission status	_selectedStatusCode string	Picker dialog (StatusEntries)	Filter N/A; Sort N/A	Codes "0", "1", "9" or labels
Notes	_commentaireCtrl TextEditingController	TextArea (multi-line)	Filter N/A; Sort N/A	Optional

Child modal dialogs (from the form):

1. **Client Picker** — Opens AlertDialog titled "Requester company", width 520px
 - Autocomplete search TextField (debounce 250ms)
 - Separated ListView with _AutocompleteEntry<int> entries (id + label)
 - Loading indicator during search
 - Retrieved from ClientService.getClientSummaries() filtered by query
 - Fallback for current client if outside the list (fallback label "Client #id (out of list)")
2. **Contact Picker** — Opens AlertDialog titled "Requester person", width 520px
 - Requires selected client first (otherwise validation "Select a requester company first")
 - Search TextField + filters by displayName, email, phone, mobile (case-insensitive contains)
 - ListView contacts filtered contactOptions
 - Visible fields: displayName (title) + subtitle (email • phone • mobile)
 - Cache per clientId in _contactsCache
3. **Interpreter Picker** — Opens AlertDialog titled "Interpreter", width 520px
 - Autocomplete search TextField (debounce 250ms)
 - ListView entries with MissionService.getInterpretes(query)
 - Fallback for current interpreter if outside the list
4. **Language Picker** — Opens AlertDialog titled "Language", width 520px
 - Autocomplete search TextField (debounce 250ms)
 - ListView entries with LanguageService.getLanguages(query)
 - Displays label (displayName); value = ref or displayName
5. **Status Picker** — Opens AlertDialog via _showEntryPicker<String>()
 - Title "Mission status"
 - Entries = _statusCodes mapped to labels via widget.labelForStatus(code)
 - Search TextField by label or code

6. **Date Picker** — Uses standard Flutter `showDatePicker()`
 - Display format dd/MM/yyyy
 - Range [-5 years, +5 years]
7. **Time Picker** — Uses standard Flutter `showTimePicker()` (implicit via tapped Time field)
 - Display format HH:mm
8. **Date Range Picker** — Uses standard Flutter `showDateRangePicker()`
 - Initializes range from last selection or current month
 - Range [-5 years, +5 years]

4.1.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Select mission type	Click FilterChip	N/A	Adds/removes type from _selectedMissionTypes (mono-select: clear then add)	Chip selected state reflected
Enter label	Edit _labelCtrl	N/A	Free text stored	Direct display
Choose date	Click _dateCtrl (readOnly)	N/A	Opens showDatePicker; sets _selectedMissionDate	Date displayed in dd/MM/yyyy format in field
Choose time	Click _heureCtrl (readOnly)	N/A	Opens showTimePicker; sets _selectedMissionTime	Time displayed in HH:mm format in field
Enter duration	Edit _dureeCtrl	N/A	Parse input (3h, 2h30, 2:30, 150 minutes)	Direct display; error if invalid format on submit
Open Client Picker	Click _showClientPicker (button/field)	N/A	Opens AlertDialog + runSearch(""); displays suggestions	Selected client → _selectedClientId, _clientCtrl.text updated
Open Contact Picker	Click _showContactPicker	Selected client	Loads client contacts; opens AlertDialog	Selected contact → _selectedContactId, _contactCtrl.text updated
Open Interpreter Picker	Click _showInterpreterPicker	N/A	Opens AlertDialog; displays MissionService.getInterpreters()	Selected interpreter → _selectedInterpreterId, _interpreterCtrl.text updated
Open Language Picker	Click _showLanguagePicker	N/A	Opens AlertDialog; displays LanguageService.getLanguages()	Selected language → _selectedLanguageRef, _langueCtrl.text updated
Open Status Picker	Click _showStatusPicker	N/A	Opens AlertDialog via _showEntryPicker; displays _statusCodes	Selected status → _selectedStatusCode, _statusCtrl.text updated

Action	Trigger	Precondition	Effect	Feedback
Search client	Edit Client Picker TextField	Picker open	Debounce 250ms → run- Search(query) → ClientSer- vice.getClientSummaries()	Filtered results displayed; loading indicator
Search contact	Edit Contact Picker TextField	Picker open	Client-side filter on displayName, email, phone, mobile	Filtered results displayed
Search interpreter	Edit Interpreter Picker TextField	Picker open	Debounce 250ms → run- Search(query) → MissionSer- vice.getInterpretes()	Filtered results displayed; loading indicator
Search language	Edit Language Picker TextField	Picker open	Debounce 250ms → run- Search(query) → LanguageSer- vice.getLanguages()	Filtered results displayed; loading indicator
Manage compa- nies/contacts	Click "Manage companies" link (optional)	N/A	Navigate /requesters; after return reload lookups	ClientSummaries + ContactService reloaded
Submit creation	Click "Validate" (or "Submit") button	Validation successful	Call _handleSubmit() → MissionSer- vice.addMissionMap(payload)	SnackBar "Mission created", display summary (modal) or close
Submit update	Click "Validate" button	Validation successful	Call _handleSubmit() → MissionSer- vice.updateMissionMap(payload)	SnackBar "Mission updated", dialog closes
Cancel creation/update	Click "Cancel"	N/A	Navigator.pop(false) or onCancel callback	Dialog closes without changes
Edit created mission	Click "Edit" button (post-creation)	Summary visible	Toggle _showCreatedSummary displays form again	Form displayable for editing
Duplicate created mission	Click "Duplicate" button	Summary visible	Call _startDuplicatedMissionEntry(); resets form shape (keeps values except ID)	Form cleared, ready for new entry
Create new after creation	Click "New mission" button	Summary visible	Call _startNewMissionEntry(); fully resets	Empty form, ready for new entry

4.1.4 States & edge cases

- **Validation failed:** Red message at top of form (e.g.: "Select a requester person"); Submit button remains active (allows correction)
- **Required fields not filled:**
 - Contact: Required

- Client: Required
- Interpreter: Required
- Language: Required (at least produit_ref or _langueCtrl non-empty)
- Date: Required
- Time: Optional (default 00:00 if empty)
- Duration: Optional (default NULL if empty)
- **Submission in progress:** `_submitting=true` → Submit button disabled, TextFields partially disabled, loading indicator
- **Server error:** Friendly Snackbar message (e.g.: "A similar mission already exists" if duplicate detected server-side)
- **Created successfully:**
 - Modal: Closes automatically after 500ms debounce for feedback
 - Embedded: Displays snapshot summary with 3 buttons (Edit, Duplicate, New) + green success message
- **Empty pickers:** Displays "No [company/interpreter/language] available" if lookup empty or request fails
- **Client changed:** Previous contacts erased; contact picker locked until new client confirmed
- **Contact not found in new query:** Erases `_selectedContactId`; contact picker resets
- **Embedded vs modal mode:**
 - Embedded: Displays summary after creation, Edit/Duplicate/New buttons
 - Modal: Closes dialog, `onSuccess(isCreation)` callback that triggers table reload
- **Flexible time format:** "14", "14h", "14:30", "14h30" all accepted and normalized HH:mm
- **Flexible duration format:** "2", "2h", "2:30", "2h30", "150" (min) all accepted and converted to minutes
- **Creation without time/duration:** Valid; default time 00:00, end = start
- **Empty interpreter search:** Displays all interpreters (limit 1000, compact limit 200)
- **Empty language search:** Displays all languages (limit 1000, compact limit 200)

4.1.5 Important business rules

- **Interpreter required:** Every mission must be assigned to 1 interpreter
- **Client + Contact required:** Every mission must have requester company + person
- **Language required:** At least produit_ref (product/language ref) must be provided
- **Planning dates:** `debutmission = datemission + heuredebutmission`; `finmission = debutmission + dureemission`
- **Multi mission types:** Can have 1+ types (Judicial Court, Translation, Interpretation)
- **Default status:** If no status selected, default "1" (Validated)
- **Implicit creator:** If creation, `creator_id` = current `AuthManager.userId`
- **Implicit modifier:** If update, `modifier_id` = current `AuthManager.userId`
- **Final validation:** Client/Contact/Interpreter/Language/Date = all non-nullable on submission
- **Duplication:** Copies all fields except ID (allows identical entry for recurring missions)

4.1.6 UI dependencies

- **Services:** `MissionService` (add/update/getInterpretes), `ClientService`, `ContactService`, `LanguageService`, `AuthManager`
- **Models:** `MissionApiResult`, `ClientSummary`, `ContactInfo`
- **Widgets:** `SelectableText`, `TextField`, `FilterChip`, `AlertDialog`, `showDatePicker`, `showTimePicker`, `LinearProgressIndicator`
- **Parent state:** `_MissionsTablePageState` (passes format/parse/decode helper functions)

4.1.7 Notes for the designer (to preserve)

- **Modal aspect ratio:** 900px × 720px (portrait-friendly, fit 1024×768+)
- **Embedded in sidebar:** Full width Card in scrollable panel
- **Form sections:** Top-to-bottom logical groups (type+label | planning | actors | details | actions)
- **Spacing:** 20px sections, 8-12px fields, 16px inter-groups
- **Error container:** Light red (E8F5E9 → red inverse) with `check_circle` icon
- **Summary container:** White background, D0D5DD border, 20px padding

- **Summary fields:** 260px width wrap layout for multi-resolution reusability
- **Autocomplete debounce:** 250ms to avoid server spam
- **Loading states:** LinearProgressIndicator minHeight=2 for discrete feedback
- **ReadOnly fields:** _dateCtrl, _heureCtrl, _statusCtrl readOnly=true to force picker usage

Figure 4.1: missions-form-create

4.2 Mission Form — Client Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Child modal dialog opened from mission form (click "Requester company" button/field)

Business purpose: Select the requester company (client) for a mission. Lookup with autocomplete. Loads contacts associated with the client after selection.

API endpoints: - POST `api/get_clients.php` (via `ClientService.getClientSummaries`) — client lookup with optional query, limit

4.2.1 Layout & zones

- **AlertDialog:** Title "Requester company", width 520px, minSize content
- **Autocomplete TextField:** autofocus=true, hintText "Search a company", magnifier prefixIcon, isDense, outline border
- **Loading indicator:** LinearProgressIndicator (minHeight 2) if loading
- **ListView:** Separated Divider(height=1), ListTile [title=label, subtitle="ID xyz"], onTap selection
- **Empty state:** Center + Text "No company [available|found for this search|unavailable]"

4.2.2 Data displayed (columns / fields)

Field	Source	Format	Notes
Title (label)	<code>_AutocompleteEntry.label</code>	Text	Company name or "Client #id"
Subtitle	<code>_AutocompleteEntry.value</code>	"ID xyz"	Rowid identifier

4.2.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Search company	Edit TextField	Picker open	Debounce 250ms → run- Search(query) → ClientSer- vice.getClientSummaries(query)	Filtered results displayed, loading indicator
Select company	Click ListTile	N/A	Sets <code>selectedEntry;</code> <code>Navigator.pop()</code>	Picker closes, return to form
Cancel	Click "Cancel" TextButton	N/A	<code>Navigator.pop()</code>	Picker closes without selection

4.2.4 States & edge cases

- **Initial loading:** Displays all clients (query='')
- **No results:** "No company found for this search"
- **Search API unavailable:** "Company search unavailable"
- **Client out of list (fallback):** If `_selectedClientId` exists but not in results, adds fallback entry "Client #id (out of list)" at top
- **Debounce request cancellation:** If new request launched before previous response, old results ignored via `requestId`

4.2.5 Important business rules

- **Mono client selection:** One client per mission (no multi-select)
- **Company influences contacts:** After client selection, immediately reloads the client's contacts
- **Cleared contacts on change:** If user changes client after selecting contact, contacts are reset

4.2.6 UI dependencies

- **Services:** `ClientService.getClientSummaries`
- **Models:** `ClientSummary`, `_AutocompleteEntry`
- **Widgets:** `AlertDialog`, `TextField`, `LinearProgressIndicator`, `ListView`, `ListTile`, `Divider`

4.3 Mission Form — Contact Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Child modal dialog opened from mission form (click "Requester person" button)

Business purpose: Select the contact (requester person) for a mission. Lists contacts of the selected client. Client-side search by `displayName`, email, phone, mobile.

API endpoints: - GET `api/get_contacts.php` (via `ContactService.getContactsForClient`) — contacts for a given client

AMI - Assistance missions interprètes

Interprètes Missions Facturation Tiers Admin

Bienvenue Anna FALL

Nouvelle mission

Les champs avec astérisque (*) sont obligatoires

Libellé

Planification

Date de la mission * (JJ/MM/AAAA) Heure de début (HH:MM) Durée (minutes)

Informations demandeur [Gérer les demandeurs](#)

Société demandeur *

Personne demandeur *

Mission

Interprète * Langue * (ref produit)

Commentaires

Statut mission

Créer

Figure 4.2: missions-form-client-picker

4.3.1 Layout & zones

- **AlertDialog:** Title "Requester person", width 520px, minSize content
- **Search TextField:** autofocus, hintText "Search a contact", magnifier prefixIcon, isDense, outline border
- **Flexible ListView:** shrinkWrap, ListTile [title=displayName, subtitle="email • phone • mobile"]
- **Empty state:** Center + Text "No contact found for this search" or "No contact available for this company"

4.3.2 Data displayed (columns / fields)

Field	Source	Format	Notes
Title	ContactInfo.displayName	Text	"FirstName LastName"
Subtitle	ContactInfo.email/phone/mobile	email • phone • mobile	Dynamic combination, separated by " • "

4.3.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Filter contacts	Edit TextField	Picker open	Client-side filter: display-Name/email/phone/mobile contains query (case-insensitive)	Filtered list displayed immediately

Action	Trigger	Precondition	Effect	Feedback
Select contact	Click ListTile	N/A	Sets selectedContact; Navigator.pop()	Picker closes, contact updated in parent
Cancel	Click TextButton	N/A	Navigator.pop() without change	Picker closes

4.3.4 States & edge cases

- **Client not selected first:** Validation “Select a requester company first”; picker does not open
- **No contact for this company:** Snackbar “No contact available for this company.”
- **Empty contacts after search:** “No contact found for this search.”
- **Contact cache per clientId:** _contactsCache[clientId] stores to optimize reloads
- **Contact resolution from input:** If user types contact name in TextField without opening picker, _resolveContactSelectionFromInput() looks for match in _contactOptions or reloads if needed

4.3.5 Important business rules

- **Mono-select contact:** One contact per mission
- **Contacts filtered by client:** Always contacts of currently selected client
- **Search fallback:** If contact typed manually not found, requests full reload of client contacts

4.3.6 UI dependencies

- **Services:** ContactService.getContactsForClient
- **Models:** ContactInfo
- **Widgets:** AlertDialog, TextField, ListView, ListTile, Divider

4.4 Mission Form — Interpreter Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Child modal dialog opened from mission form (click “Interpreter” button)

Business purpose: Select the interpreter (assign a mission). Server-side lookup with autocomplete via MissionService.

API endpoints: - POST api/get_interpretes.php (via MissionService.getInterpretes) — interpreter lookup with optional query, limit

4.4.1 Layout & zones

- **AlertDialog:** Title “Interpreter”, width 520px
- **Autocomplete TextField:** autofocus, hintText “Search an interpreter”, magnifier prefixIcon, isDense, outline border
- **Loading indicator:** LinearProgressIndicator during search
- **ListView:** ListTile [title=display_name, subtitle=“ID xyz”]
- **Empty state:** Center + Text message

4.4.2 Data displayed (columns / fields)

AMI - Assistance missions interprètes

Interprètes Missions Facturation Tiers Admin

Bienvenue Anna FALL

Nouvelle mission

Les champs avec astérisque (*) sont obligatoires

Libellé

Planification

Date de la mission * (JJ/MM/AAAA) Heure de début (HH:MM) Durée (minutes)

Informations demandeur

Société demandeur * Personne demandeur *

Mission

Interprète * Langue * (ref produit)

Commentaires

Statut mission

Brouillon

Créer

Figure 4.3: missions-form-contact-picker

Field	Source	Format	Notes
Title	entry['display_name']	Text	"FirstName LastName" or "Interpreter #id"
Subtitle	entry['id']	"ID xyz"	Identifier

4.4.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Search interpreter	Edit TextField	Picker open	Debounce 250ms → run-Search(query) → MissionService.getInterpretes(query)	Filtered results, loading indicator
Select interpreter	Click ListTile	N/A	Sets selectedEntry; Navigator.pop()	Picker closes
Cancel	Click TextButton	N/A	Navigator.pop()	Picker closes

4.4.4 States & edge cases

- **No interpreter available:** "No interpreter available."
- **Search unavailable:** "Interpreter search unavailable."
- **Interpreter out of list (fallback):** If `_selectedInterpreterId` exists but absent from results, adds fallback "Interpreter #id (out of list)"

- **Request debounce:** Cancellation of old request if new one launched

4.4.5 Important business rules

- **Mono-select interpreter:** One interpreter per mission
- **Interpreter required:** Validation imposes selection

4.4.6 UI dependencies

- **Services:** MissionService.getInterpretes
- **Models:** _AutocompleteEntry
- **Widgets:** AlertDialog, TextField, ListView, ListTile, LinearProgressIndicator

Figure 4.4: missions-form-interpreter-picker

4.5 Mission Form — Language Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Child modal dialog opened from mission form (click "Language" button)

Business purpose: Select the language/service product for a mission. Server-side lookup with autocomplete via LanguageService.

API endpoints: - GET `api/get_languages.php` (via `LanguageService.getLanguages`) — language lookup with optional query, limit

4.5.1 Layout & zones

- **AlertDialog:** Title "Language", width 520px

- **Autocomplete TextField:** autofocus, hintText "Search a language", magnifier prefixIcon, isDense, outline border
- **Loading indicator:** LinearProgressIndicator during search
- **ListView:** ListTile [title=label (displayName)]
- **Empty state:** Center + Text message

4.5.2 Data displayed (columns / fields)

Field	Source	Format	Notes
Title	option.displayName	Text	E.g.: "French", "English"

4.5.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Search language	Edit TextField	Picker open	Debounce 250ms → run- Search(query) → LanguageSer- vice.getLanguages(query)	Filtered results, loading indicator
Select language	Click ListTile	N/A	Sets selectedEntry (value=ref or displayName); Navigator.pop()	Picker closes
Cancel	Click TextButton	N/A	Navigator.pop()	Picker closes

4.5.4 States & edge cases

- **No language available:** "No language available."
- **Search unavailable:** "Language search unavailable."
- **Picker reopen:** Resets _languageSearchEntries to load fresh (avoids stale cache)

4.5.5 Important business rules

- **Mono-select language:** One product/language per mission
- **Language required:** Validation imposes entry
- **Ref or Label:** Stores ref if non-empty, otherwise displayName

4.5.6 UI dependencies

- **Services:** LanguageService.getLanguages
- **Models:** _AutocompleteEntry
- **Widgets:** AlertDialog, TextField, ListView, ListTile, LinearProgressIndicator

4.6 Mission Form — Status Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Child modal dialog opened from mission form (click "Mission status" button/field)

Business purpose: Select the workflow status of a mission (Draft, Validated, Cancelled). Display codes and labels.

API endpoints: N/A (local _statusCodes lookup)

The screenshot shows the 'Nouvelle mission' (New mission) form in the AMI - Assistance missions interprètes application. The form is titled 'Nouvelle mission' and includes a note: 'Les champs avec astérisque (*) sont obligatoires'. The form is divided into several sections:

- Planification:** Includes fields for 'Date de la mission * (JJ/MM/AAAA)', 'Heure de début (HH:MM)', and 'Durée (minutes)'.
- Informations demandeur:** Includes fields for 'Société demandeur *' and 'Personne demandeur *'. There is a link 'Gérer les demandeurs'.
- Mission:** Includes a dropdown for 'Interprète *', a dropdown for 'Langue * (ref produit)', and a text area for 'Commentaires'.
- Statut mission:** Includes a dropdown for 'Statut mission' with the option 'Brouillon'.

A 'Créer' button is located at the bottom right of the form.

Figure 4.5: missions-form-language-picker

4.6.1 Layout & zones

- **AlertDialog:** Title "Mission status", width 520px
- **Search TextField:** autofocus, hintText "Search", magnifier prefixIcon, isDense
- **ListView:** ListTile [title=label, subtitle=code]
- **Empty state:** Center + Text "No status available"

4.6.2 Data displayed (columns / fields)

Field	Source	Format	Notes
Title	_statusCodes mapped to labelForStatus	Text	"Draft", "Valid", "Cancelled"
Subtitle	_statusCodes	Code (0, 1, 9)	Internal identifier

4.6.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Search status	Edit TextField	Picker open	Client-side filter on label (contains, case-insensitive)	Filtered list
Select status	Click ListTile	N/A	Sets <code>selectedEntry;</code> <code>Navigator.pop()</code>	Picker closes, status updated

Action	Trigger	Precondition	Effect	Feedback
Cancel	Click TextButton	N/A	Navigator.pop()	Picker closes

4.6.4 States & edge cases

- **No status available:** Message if _statusCodes empty (should not happen in production)
- **Default status:** If creation, default "1" (Valid); if edit, loads current status

4.6.5 Important business rules

- **Mono-select status:** One status per mission
- **Valid statuses:** 0=Draft, 1=Validated, 9=Cancelled
- **Status workflow ≠ invoice:** Distinct from interpreter/client billing statuses

4.6.6 UI dependencies

- **Models:** _statusCodes list
- **Widgets:** AlertDialog, TextField, ListView, ListTile, _showEntryPicker

The screenshot shows the 'Nouvelle mission' form in the AMI - Assistance missions interprètes application. The form is titled 'Nouvelle mission' and includes a note: 'Les champs avec astérisque (*) sont obligatoires'. The form is divided into several sections: 'Type de mission' with a dropdown menu showing 'Libellé'; 'Planification' with fields for 'Date de la mission * (JJ/MM/AAAA)', 'Heure de début (HH:MM)', and 'Durée (minutes)'; 'Informations demandeur' with fields for 'Société demandeur *' and 'Personne demandeur *'; 'Mission' with fields for 'Interprète *' and 'Langue * (ref produit)'; 'Commentaires' with a text area; and 'Statut mission' with a dropdown menu showing 'Brouillon'. A 'Créer' button is located at the bottom right of the form.

Figure 4.6: missions-form-status-picker

4.7 Delete Mission Confirmation — [lib/pages/missions_table_page.dart](#)

Route / access: Modal confirmation dialog opened from table (click delete button on row)

Business purpose: Confirm cancellation/deletion of a mission before executing. Hard-delete via soft-delete (status=9).

API endpoints: - POST `api/delete_mission_interpreter.php` — soft-delete mission (set status=9)

4.7.1 Layout & zones

- **AlertDialog:** Title "Cancel the mission?", short content, 2 action buttons
- **Content text:** "The mission will be set to Cancelled status."

4.7.2 Data displayed (columns / fields)

N/A (minimal display)

4.7.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Confirm cancellation	Click "Confirm"	N/A	Call <code>MissionService.deleteMission(rowId)</code> → POST <code>/api/delete_mission_interpreter.php</code>	SnackBar "Mission cancelled", table reload
Refuse cancellation	Click "Cancel"	N/A	<code>Navigator.pop(ctx, false)</code>	Closes dialog, no change

4.7.4 States & edge cases

- **Server error:** SnackBar "Deletion failed"; mission remains visible
- **Non-mounted after dialog:** Check `!mounted` before UI update

4.7.5 Important business rules

- **Soft-delete:** Mission not physically deleted, status=9 (Cancelled); remains visible in table but non-billable
- **"Non-billable" badge:** Cancelled missions display this badge in table

4.7.6 UI dependencies

- **Services:** `MissionService.deleteMission`
- **Widgets:** `AlertDialog`, `TextButton`, `ElevatedButton`

4.8 Mission Creation Success Summary — [lib/pages/missions_table_page.dart](#)

Route / access: Displayed after successful mission creation in embedded form ("New mission" sidebar)

Business purpose: Display recap of created mission with key fields. Suggest quick actions (Edit, Duplicate, New mission).

API endpoints: N/A (local display of created snapshot)

4.8.1 Layout & zones

- **Container:** White background, D0D5DD border, 20px padding, 10px borderRadius
- **Success message box:** Light green background (E8F5E9), 81C784 border, 12px padding, contains `check_circle` icon + success text

AMI - Assistance missions interprètes

Interprètes

Missions

Facturation

Tiers

Admin

Bienvenue Anna FALL

Nouvelle mission

Les champs avec astérisque (*) sont obligatoires

Nouvelle mission

Tribunal judiciaire

Traduction

Interprétariat

Libellé

Planification

Date de la mission * (JJ/MM/AAAA)

Heure de début (HH:MM)

Durée (minutes)

Informations demandeur

Société demandeur *

Personne demandeur *

Gérer les demandeurs

Mission

Interprète *

Langue * (ref produit)

Commentaires

Statut mission

Brouillon

Créer

Figure 4.7: missions-delete-confirm

- **Title:** "Mission created"
- **Action buttons:** Row [Edit, Duplicate, New mission]
- **Field wrap:** Grid display 260px × N snapshot fields

4.8.2 Data displayed (columns / fields)

Field	Label	Source	Format
Mission ref.	"Mission ref."	snapshot['reference_devis']	Text
Status	"Status"	snapshot['mission_status']	Text
Mission type	"Mission type"	mapped labelForStatus	
Label	"Label"	snapshot['label']	Text
Mission date	"Mission date"	snapshot['datemission']	yyyy-MM-dd
Start time	"Start time"	snapshot['heuredebutmission']	HH:mm
Duration	"Duration"	snapshot['dureemission']	Minutes
Company	"Requester company"	snapshot['client_name']	Text
Contact	"Requester person"	snapshot['contact_name']	Text
Interpreter	"Interpreter"	snapshot['interpreter_name']	Text
Language	"Language"	snapshot['produit_ref']	Text
Notes	"Notes"	snapshot['commentaires']	Text (multi-line)

4.8.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Edit	Click "Edit"	Summary visible	Toggle _showCreatedSummary displays form	User can edit created mission
Duplicate	Click "Duplicate"	Summary visible	Call _startDuplicatedMission(); resets except ID	User can create similar mission
New	Click "New mission"	Summary visible	Call _startNewMissionForNewEntry(); fully resets	Empty form ready for new entry

4.8.4 States & edge cases

- **Embedded mode only:** `_showCreatedSummary=true && widget.embedded && _createdMissionSnapshot != null`
- **Limited snapshot:** Contains only entered fields + API return (ref, id)
- **Optional success message:** Displayed if `_successMessage` populated

4.8.5 Important business rules

- **Duplication:** Keeps all fields except ID for quick recurrence (recurring missions)
- **UX display:** Positive feedback before empty form, encourages next action

4.8.6 UI dependencies

- **Models:** `_createdMissionSnapshot Map<String, dynamic>`
- **Widgets:** Container, Icon (check_circle), OutlinedButton, FilledButton, Row, Wrap

4.9 Column Picker Dialog — [lib/pages/missions_table_page.dart](#)

Route / access: Modal dialog opened from table (click "Columns to display" button)

Business purpose: Allow user to choose which columns to display in the missions table. Local persistence via `_visibleColumns` map.

API endpoints: N/A (UI local only)

4.9.1 Layout & zones

- **AlertDialog:** Title "Visible columns", content `SingleChildScrollView`
- **Column:** `MainAxisSize.min`, `crossAxisAlignment start`
- **SwitchListTile** × N: One for each optional column
- **TextButton "Show all":** Align `centerLeft`

4.9.2 Data displayed (columns / fields)

Column	Label	Visible by default	Notes
refmission	"Mission Id."	false	rowid column
telephone	"Requester Tel."	true	contact phone
mobile	"Requester Mobile"	true	contact phone_mobile
typeMission	"Mission type"	true	mission_types

Column	Label	Visible by default	Notes
facture	"Interpreter invoice"	true	billed_status
statut	"Mission status"	true	mission_status
factureClient	"Client invoice"	true	client_invoice_number + badge
createur	"Created by"	true	creator_name
datecrea	"Creation date"	true	date_creation
dateModif	"Modification date"	false	date_modification (optional)
modifiePar	"Updated by"	false	updated_by (optional)

4.9.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Toggle column visibility	Click SwitchListTile	Dialog open	Toggle _visibleColumns[style] via setLocalState	Switch UI reflects state
Show all columns	Click "Show all"	Dialog open	Toggle all _visibleColumns[*] = true	All switches ON
Apply changes	Click "Apply"	Dialog open	Call setState() parent; Navigator.pop()	Dialog closes, table re-filters columns
Cancel	Click "Cancel"	Dialog open	Navigator.pop()	Dialog closes, local changes lost (setLocalState not committed)

4.9.4 States & edge cases

- **No column selected:** Technically possible but bad UX — no dialog-side validation
- **Actions column always displayed:** "Actions" checkbox absent from picker (implicitly always visible)
- **setLocalState vs setState:** setLocalState for live preview without commit; setState parent on Apply

4.9.5 Important business rules

- **Persistent visibility (session):** _visibleColumns kept during session (not localStorage/prefs)
- **Always visible key columns:** "ref", "langue", "typeMission", "datemission", "heuredebut", "duree", "societe", "demandeur", "interprete", "label", "statut", "actions"

4.9.6 UI dependencies

- **Widgets:** AlertDialog, SingleChildScrollView, Column, SwitchListTile, TextButton, FilledButton

AMI - Assistance missions interprètes

InterprètesMissionsFacturationTiersAdmin

Bienvenue Anna FALL

Nouvelle mission

Les champs avec astérisque (*) sont obligatoires

Nouvelle mission

Tribunal judiciaireTraductionInterprétariat

Libellé

Planification

Date de la mission * (JJ/MM/AAAA)Heure de début (HH:MM)Durée (minutes)

Informations demandeur

Société demandeur *Personne demandeur *

Mission

Interprète *Langue * (ref produit)

Commentaires

Statut missionBrouillon

Créer

Figure 4.8: missions-creation-success

AMI - Assistance missions interprètes

InterprètesMissionsFacturationTiersAdmin

Bienvenue Anna FALL

Nouvelle mission

Les champs avec astérisque (*) sont obligatoires

Nouvelle mission

Tribunal judiciaireTraductionInterprétariat

Libellé

Planification

Date de la mission * (JJ/MM/AAAA)Heure de début (HH:MM)Durée (minutes)

Informations demandeur

Société demandeur *Personne demandeur *

Mission

Interprète *Langue * (ref produit)

Commentaires

Statut missionBrouillon

Créer

Figure 4.9: missions-column-picker

Chapter 5

4. Quotes and company settings

5.1 Company settings — [lib/pages/company_info_page.dart](#)

Route / access: Conditional access: visible only for admin users (check `userRights.isAdmin()`). From navigation menu or admin menu. Parameter: `userRights: UserRights`.

Business purpose: Allow the administrator to centralize and maintain the legal/contact information of the issuing company (SIRET, contact info, website, logo). This data is used in all generated PDFs (quotes, invoices, etc.) and in [api/get_company_info.php](#) → [api/get_company_bank_accounts.php](#).

API endpoints: - [api/get_company_info.php](#) — retrieve existing configuration + bank accounts - [api/update_company_info.php](#) — save modifications - [api/get_company_bank_accounts.php](#) — list bank accounts (preview display only)

5.1.1 Layout & zones

AppBar: - Title: « Company settings » - Refresh button (icon) — reloads data from [api/get_company_info.php](#)

Scrollable area (body): - **Error message** (conditional): Yellow banner if partial loading - « **Identity** » **section:** - Text field: Company name (required) - Text field: Logo URL (optional, URL) - « **Address** » **section:** - Text field: Address line 1 (required) - Text field: Address line 2 (optional) - Row (flex): Postal code (required) + City (required) - « **Legal & contact information** » **section:** - Text field: Siret (required) - Text field: Phone (optional) - Text field: Email (optional, email validation) - Text field: Website (optional, URL validation) - **Preview card** (`_buildPreviewCard()`): - Apartment icon (72×72 px, gray background) - On the right: « Issuer » gray label, company name gray/bold, addresses, SIRET, contact - If bank accounts present: « Bank details » section with label, bank, holder, IBAN, BIC

Bottom: - Elevated button « Save » — saves via [update_company_info.php](#) - Brand footer (BrandFooter)

5.1.2 Data displayed

Field	Source	Format	Sort/Filter	Notes
Name	llx_const MAIN_INFO_SOCIETE_NOM	Text max 150	—	Required, default « Planet Traduction »
Address 1	llx_const MAIN_INFO_SOCIETE_ADRESSE*	Text max 200	—	Required
Address 2	llx_const MAIN_INFO_SOCIETE_ADRESSE2*	Text max 200	—	Optional
Postal code	llx_const MAIN_INFO_SOCIETE_CP*	Text max 30	—	Required

Field	Source	Format	Sort/Filter	Notes
City	llx_const MAIN_INFO_SOCIETE_VILLE*	Text max 150	—	Required
SIRET	llx_const MAIN_INFO_SOCIETE_SIRET	Text max 50	—	Required, fallback « 91282415800014 »
Phone	llx_const MAIN_INFO_SOCIETE_TEL	Text max 50	—	Optional
Email	llx_const MAIN_INFO_SOCIETE_MAIL	Text max 150	—	Optional, FIL- TER_VALIDATE_EMAIL validation
Website	llx_const MAIN_INFO_SOCIETE_WEB	Text max 200	—	Optional, URL validation
Logo URL	llx_const MAIN_INFO_SOCIETE_LOGO_URL*	Text max 300	—	Optional, relative or absolute, URL validation if starts with ://
Bank accounts (preview)	llx_bank_account (main = courant DESC, completeness DESC)	Text display	—	Read-only, sorted by default bank, label, IBAN, BIC, holder

5.1.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Check access	Page init/build	—	Validation <code>userRights.isAdmin()</code>	If non-admin, displays message « You do not have rights... »
Load config	Page init <code>_load()</code> or refresh btn	—	GET api/get_company_info.php	Loading spinner, populate form, if error: fallback + yellow error banner
Edit field	Edit in <code>TextFormField</code>	—	Update local controller	Preview card updates in real time (setState)
Validate email	onChange on email field	Non-empty email	<code>FILTER_VALIDATE_EMAIL</code> validation	Alert « Invalid email » if failed
Validate URL	onChange on website field	Non-empty website	<code>FILTER_VALIDATE_URL</code> validation	Error « Invalid website URL » if failed
Validate logo URL	onChange on logoUrl field	Non-empty logoUrl AND contains ://	<code>FILTER_VALIDATE_URL</code> validation	Error « Invalid logo URL » if problem
Submit form	Click « Save » btn	All required fields valid + state != saving	FormKey validation, POST update_company_info.php	Btn disabled, loading spinner, success/error SnackBar
Refresh data	Click refresh btn (appbar)	—	<code>Restart_load()</code>	Refresh spinner appbar, reload form

5.1.4 States & edge cases

- **Initial loading state:** `_loading=true`, displays central `CircularProgressIndicator`
- **Non-admin state:** Page displays « You do not have rights to access this page »
- **Loading error state:** `_error != null`, displays yellow warning banner, fallback `CompanyInfo.fallback()` used
- **Saving state:** `_saving=true`, « Save » btn disabled + spinner, all inputs disabled (optional but consistent UX)
- **Validation fails:** `FormKey` validation fail, btn does not save, specific errors displayed under fields
- **Invalid email/URL:** Client-side + server-side validation (PHP `filter_var`), rejection if incorrect format
- **Partial logo URL:** If no protocol (`http://`, `https://`), accepted (relative path possible)
- **No change:** Submission even if no change, server updates anyway (idempotent)
- **Reactive preview:** Preview card updates instantly on each keystroke (no debounce) via `setState` `onChanged`

5.1.5 Important business rules

- **Admin access:** Only admin (`UserRights.isAdmin()`) can access
- **Required fields:** Name, Address 1, Postal code, City, SIRET
- **Validation:** Email regex `FILTER_VALIDATE_EMAIL`, URL `FILTER_VALIDATE_URL`, logo URL if absolute
- **SIREN extraction:** From SIRET, extracts first 9 digits → stores separately in `MAIN_INFO_SOCIETE_SIREN`
- **Dolibarr storage:** Data persisted in `llx_const` (Dolibarr constants table), `entity=DOLIBARR_ENTITY`
- **Fallback values:** If `getenv('DOLIBARR_ENTITY')` empty, `entity=1`; hardcoded defaults « Planet Traduction »
- **Timestamps:** Min/Max of `tms llx_const` stored as `createdAt/updatedAt` for audit
- **Bank accounts:** Automatic sorting: Banque Populaire Rives de Paris prioritized, then `courant=1`, then completeness (IBAN, BIC, holder, number)
- **IBAN composition:** If `iban_prefix` non-empty, used as-is; otherwise reconstructed from `country_iban` + `cle_iban` + `code_banque` + `code_guichet` + `number` + `cle_rib`

5.1.6 UI dependencies

- `CompanyInfoService` — fetch/update
- `UserRights` — admin rights check
- `ResponsiveHelper` — `ResponsiveContainer` for adapted spacing
- `BrandFooter` — recurring footer
- Flutter forms (`FormKey`, `TextFormField`) — client-side validation

5.1.7 Notes for the designer

- **Section structure:** Bold 16px section title, fields grouped by business domain (Identity, Address, Legal)
- **Text fields:** `InputDecoration` with label, `OutlineInputBorder` border, `isDense=true` (compact)
- **Address row:** Postal code and City side by side, equal flex, 12px gap
- **Preview card:** Light `E5E7EB` border, 12px radius, white background, miniature apartment icon, rich text (bold on name, gray on addresses)
- **Bank accounts in preview:** Optional section if `hasBankDetails`, 10px spacing before, small gray7280 text, reduced info (label, bank, IBAN, BIC)
- **Save button:** `ElevatedButton`.icon, save icon, full width, visual feedback (spinner if saving)
- **Responsive:** For mobile/tablet use `ResponsiveContainer` with dynamic spacing

5.2 Client autocomplete widget — `lib/widgets/client_autocomplete_field.dart`

Access: Reusable form widget, consumed in editing pages (quotes, invoices, etc.). Integrated as a text input component with dropdown suggestion.

Business purpose: Provide autocompleted entry of the client name with real-time search via `api/get_client_summaries` (alias `get_clients.php`). Returns the numeric ID of the selected client via `onSummarySelected` callback.

Identité

Mon de l'entreprise

Planet Traduction

URL du logo (optionnel)

Logo semi modèle espacé (1).png

Adresse

Adresse ligne 1

13 chemin des Champcueil

Adresse ligne 2

Code postal

91220

Ville

Brétigny Sur Orge

Informations légales & contact

Siret

912924158

Téléphone

0178908756

Email

contact@planettraduction.fr

Site web

https://planet-traduction.fr/

Émetteur

Planet Traduction

13 chemin des Champcueil

91220 Brétigny Sur Orge

Siret : 912924158

Tél : 0178908756

Email : contact@planettraduction.fr

Web : https://planet-traduction.fr/

Coordonnées bancaires

Compte : BP RIVES DE PARIS

Banque : BANQUE POPULAIRE RIVES DE PARIS

Titulaire : ASS PLANET TRADUCTION

IBAN : FR76 1020 7000 6724 2151 1480 210

BIC : CCBPFRPPMTG

Enregistrer

Figure 5.1: company-info

API endpoints: - [ClientService.getClientSummaries\(\)](#) — internal call to client API (probably [api/get_clients.php](#))

5.2.1 Layout & zones

Single widget: RawAutocomplete wrapper

TextField: - PrefixIcon: Icon(Icons.business) — client icon - HintText: « Select a client » (customizable) - SuffixIcon: Row with dropdown button + optional spinner

Dropdown options: - Material elevation=6, 8px radius - ConstrainedBox max 300px height, min 280px width - ListView.separated: List of names (string), light borders between items - ListTile clickable for each option

5.2.2 Data displayed

Field	Source	Format	Sort/Filter	Notes
Value (input)	widget.value prop or TextEditingController.text	Text	—	Client name entered or selected
Options	_options List	List of client names	Filter by query (contains case-insensitive)	Fetches via ClientService, max 30 items
Summaries (metadata)	_summaries List	Object {id, name, ...}	—	Kept in parallel for full ID return
Loading	_loading bool	Spinner or nothing	—	Displayed in suffix row if true

5.2.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Type text	onChange TextFormField	enabled=true	Debounce 250ms, fetch ClientService.getClientSummaries()	_options updated, _summaries synchronized, loading toggle
Show dropdown	Click dropdown btn or focus + empty text	enabled=true	Calls showOptions(), displays _options if existing or refetch	loading toggle Dropdown visible with list
Select option	Click ListTile or Enter	Option visible	Callback onSe- lected(name) + onSummarySe- lected(ClientSummary) if defined	TextField update, dropdown closes, focus lost
Submit (Enter)	onSubmitted handler	—	Callback onSubmitted if defined	Allows external validation
Network error	Fetch fails	—	_options cleared, _summaries cleared, _loading=false	onError callback called with error message
Force refresh	External control via ClientAutocompleteCon- troller.refresh()	—	Re-fetch with current query	Useful if data changed server-side

5.2.4 States & edge cases

- **Loading state:** _loading=true between keystroke and API response, spinner displayed in suffix
- **No result:** _options empty, dropdown does not display
- **Empty text, dropdown clicked:** Displays all options (limit 30)
- **Debounce:** 250ms between keystroke and API call, avoids flood
- **Request token:** _requestToken checks to ignore old responses (race condition)
- **Focus behavior:** Focus on field = displays optional dropdown; loss of focus = dropdown closes
- **Disabled:** widget.enabled=false → grayed TextField, dropdown btn disabled
- **Initial value:** widget.value propagated to TextEditingController at init and via didUpdateWidget
- **Selection from summary:** If onSummarySelected defined, searches in _summaries (s.name == selection) and returns full object with ID

5.2.5 Important business rules

- **Fuzzy search:** Case-insensitive query, contains on names (not prefix-only)
- **Limit 30 results:** Avoids UI lag, pagination not implemented in this widget
- **ClientSummary return:** Full {id, name, ...} format sent to parent callback, not just string
- **No creation:** Widget does not create clients, read-only
- **Debounce 250ms:** Standard delay, configurable if needed (const in _onChanged)
- **Race conditions:** Token system prevents displaying stale results if multiple in-flight requests

5.2.6 UI dependencies

- [ClientService](#) — getClientSummaries() API call
- Flutter RawAutocomplete — base autocomplete component

5.2.7 Notes for the designer

- **TextField:** Standard style, business icon (default color), light gray placeholder
- **Dropdown styling:** Elevation 6 (light shadow), 8px rounded corners, max 300px scrollable height
- **ListTile items:** Standard inner padding, light border between items
- **Spinner suffix:** 18×18 px, 2px stroke, default blue-gray
- **Dropdown arrow btn:** Icons.arrow_drop_down, consistent size
- **UX animation:** Smooth slide-in dropdown, no artificial delay

Chapter 6

5. Billing

6.1 BillingPage — lib/pages/billing_page.dart

Route / access: Direct access via main navigation; optional arguments `BillingPageArguments` (missions, invoiceNumber); “Billing” tab in the sidebar or section selector. **Business purpose:** Manage the complete lifecycle of client invoices: draft creation, line editing, draft management, viewing and modifying the status of issued invoices. **API endpoints:** - `get_missions_datatable.php` (POST) — retrieve missions by client/period - `reserve_client_invoice_number.php` (GET) — generate invoice no. - `save_invoice_draft.php` (POST) — save draft header - `save_invoice_draft_lines.php` (POST) — save draft lines - `get_invoice_draft_lines.php` (POST) — retrieve draft lines - `get_invoice_drafts.php` (POST) — list drafts in progress - `delete_invoice_draft.php` (POST) — delete a draft - `log_client_billing.php` (POST) — invoice creation (billing + PDF record) - `get_client_invoices.php` (POST) — list finalized invoices - `get_client_invoice_lines.php` (POST) — invoice line details - `update_client_invoice_lines.php` (POST) — line modification (for specific mission) - `update_client_invoice_status.php` (POST) — status change - `update_invoice_bank_account.php` (POST) — bank account assignment - `update_invoice_payment_term.php` (POST) — payment term assignment - `update_invoice_date.php` (POST) — creation date modification - `get_client_payment_terms.php` (GET/POST) — client payment terms - `get_company_bank_accounts.php` — company bank accounts - `get_company_info.php` — company info (for PDF)

6.1.1 Layout & zones

- **Header:** Section title (« Invoice creation », « Invoices table » or « Initiated invoices »); orientation sidebar (Draft / Invoices / Initiated invoices); responsive (mobile sidebar collapse).
- **Navigation bar:**
 - Draft: client selector (autocomplete), month, bank account, payment term, action buttons (Load missions, Create invoice, Generate PDF, Save draft, Reset).
 - Invoices: filter fields (search, month, status, client) + Refresh button + CSV export.
- **Body:**
 - Draft: billing lines table (paginated data) with inline editing (quantity, price, VAT, etc.); HT/TTC totals display.
 - Invoices: paginated invoice table or compact list (mobile); action icons (see detail, open PDF, generate PDF, cancel).
 - Initiated invoices: list of drafts with Resume / Delete buttons per item.
- **Footer / contextual actions:** Pagination + line/invoice count; HT/TTC totals; contextual delete/cancel buttons.

6.1.2 Data displayed (columns / fields) — DRAFT SECTION

Field	Source (API / model)	Format	Sort/Filter	Notes
Delete	—	Trash icon	—	Per-line row action
Mission ref.	Mission Data / MissionRef	Text, align left	—	Identifier mission group
Description	Mission / InvoiceLine	Text, editable, multi-line	—	Max 4 lines inline
VAT	InvoiceLine.tvaRate	Dropdown (% , e.g.: 20 %, 5.5 %)	—	Non editable after creation
Unit price excl. tax	InvoiceLine.unitPrice	Text, editable, 2 dec.	—	Currency format
Unit price incl. tax	InvoiceLine.unitPriceTtc	Text, read-only	—	Computed = unitPrice * (1 + VAT/100)
Discount %	InvoiceLine.discount	Text, editable, 0–100	—	Commercial discount
Qty	InvoiceLine.quantity	Text, editable, 3 dec.	—	Quantity or duration (days, hours)
Total excl. tax	InvoiceLine.totalHt	Text, selectable, align right	—	= (unitPrice - discount %) * qty
Total incl. tax	InvoiceLine.totalTtc	Text, selectable, align right	—	= totalHt * (1 + VAT/100)

6.1.3 Data displayed (columns / fields) — INVOICES SECTION

Field	Source (API / model)	Format	Sort/Filter	Notes
Invoice no.	ClientInvoiceSummary.invoiceId (blue #2563EB)	Text, clickable	☐ Primary sort	Click → opens detail
Client	ClientInvoiceSummary.clientName	Text (primary / secondary if present)	☐	Text or multi-line
Month	ClientInvoiceSummary.periodMonth / billedAt	YYMM	☐ Filter	Billing period
Total excl. tax	ClientInvoiceSummary.invoiceTotalHt / amountHt	€ formatted 2 dec.	☐	€ formatted 2 dec.
Total incl. tax	ClientInvoiceSummary.invoiceTotalTtc	€ formatted 2 dec.	☐	€ formatted 2 dec.
Status	ClientInvoiceSummary.statusCode / Label	Status Code (draft=gray, validated=green, paid=blue, cancelled=red)	☐ Filter	Dropdown editable if not paid/cancelled
Creation date	ClientInvoiceSummary.createdAt / billedAt	date YYYY-MM-DD HH:mm	☐ Sort	Clickable (date picker)
Actions	—	Icons (view, open server PDF, generate PDF, cancel)	—	Contextual depending on status

6.1.4 User actions — DRAFT SECTION

Action	Trigger	Precondition	Effect	Feedback
Select client	AutoCompleteField / Dropdown	—	Loads missions, resets fields, loads payment terms	Spinner in client field

Action	Trigger	Precondition	Effect	Feedback
Change month	DropdownButtonFormClient selected	Client selected	Recalculates missions filtered by month	Table reflects chosen month
Load missions	« Load missions » button or search icon	Client + month OK	API request, displays eligible lines (status 0/1), checks for existing draft	Spinner; dialog if draft found (Resume/Ignore)
Resume from draft	« Draft in progress » dialog — Resume button	Active draft found	Restore _lineEditors from draft, same row from DB	Panel closes, table updated
Ignore existing draft	« Draft in progress » dialog — Ignore button	Draft found, user chooses ignore	Keeps draft in DB, displays raw missions	Panel closes, mission display
Edit line (inline)	Click on cell (Description, Qty, Unit price, VAT, Disc., etc.)	Line present	Activates inline editing for cell	Text field or dropdown depending on column
Open line editor panel	Click on line (or row highlight)	Line present	Right side panel displays, loads complete form (label, qty, price, VAT, notes)	Panel slide-in from right
Delete line	Click trash icon per line	Line present	Removes from _lineEditors, removed from table	Line disappears, counter decrements
Compute totals	Auto after editing	Each quantity / price change	Recalc totalHt, totalTtc, invoice totals	Immediate display at bottom of table
Save draft	« Save draft » button or autosave (800ms)	Lines present + client OK	POST save_invoice_draft.php + save_invoice_draft_lines.php, stores draft_id/draft_key	Snackbar « Saved »; « saving » badge top-right
Reset all	« Reset all » button	—	Clear _lineEditors, client, month, draft	Confirmation dialog
Generate PDF (preview)	« Generate invoice (PDF) » button	Lines OK + client + month OK	Reserves invoice no., builds PDF, launches Printing.layoutPdf (web) or savePdfToDownloads	Spinner; confirmation snackbar or error
Create invoice	Blue « Create and save invoice » button	Lines + client + month OK, no conflict	Confirmation dialog, POST log_client_billing (creates tble_client_billed + lines)	Snackbar « Invoice NNN created »; redirect to Invoices on success

6.1.5 User actions — INVOICES SECTION

Action	Trigger	Precondition	Effect	Feedback
Load invoices	Click « Update » button or auto on switch to Invoices	—	GET <code>get_client_invoices.php</code> with applied filters	Spinner; table updates
Search invoice	Search TextField	—	Filters <code>_invoices</code> on <code>invoiceNumber</code> , <code>clientName</code> , <code>month</code> , <code>status</code> in real time	Table filters instantly, resets page 1
Filter by month	« Month » dropdown	—	Restrict invoices to chosen month or « All months »	Table filters, counter updated
Filter by client	« Client » dropdown (reloaded at each search)	—	Restrict invoices to client or « All clients »	Re-fetch with <code>client_name</code> filter
Filter by status	« All / Draft / Validated / Paid / Cancelled » chips	—	Toggle <code>_invoiceStatusFilter</code>	Table filters immediately
Paginate (list)	« First / Prev / Next / Last page » buttons or « Page X/Y »	Loaded invoices	Paginated navigation (<code>_invoiceDisplayPage</code>)	Table scroll to top; current range display
Change rows/page	« 25 / 50 / 100 / 250 » dropdown	—	Adjust <code>_invoiceDisplayPageSize</code>	Reset page 1; table re-paginates
Open invoice detail	Click on table row (or eye button)	Invoice present	Detail panel displays (right side panel or mobile modal); loads lines via <code>get_client_invoice_lines.php</code>	Panel open, spinner if lines loading
Change invoice status	« Draft / Validated / Paid / Cancelled » dropdown in detail	Invoice not locked (status $\neq$ paid/cancelled)	Optimistic update, POST <code>update_client_invoice_status.php</code>	Status badge toggles; spinner overlay if update in progress
Modify creation date	Click on date (calendar icon) in detail	Invoice not locked	Date + time picker (<code>showDatePicker</code> + <code>showTimePicker</code>)	Picker dialog opens
Modify bank account	Dropdown in detail	Invoice not locked + accounts available	Optimistic update, POST <code>update_invoice_bank_account.php</code> ; updates notes	Dropdown toggle; spinner overlay
Modify payment term	Dropdown in detail	Invoice not locked + terms loaded	Optimistic update, POST <code>update_invoice_payment_term.php</code>	Dropdown toggle; spinner
Edit invoice lines (mission-specific)	Click on mission ref in « Associated missions » table	Invoice not locked	Detail panel re-displays with mission lines + editable form	Inline edit, Reset/Save buttons
Save line editing	« Save lines » button (in mission detail)	Lines modified + invoice not locked	POST <code>update_client_invoice_lines.php</code> ; return to full-invoice detail	Snackbar « Lines updated »; panel refreshes
Generate PDF (finalized invoice)	« Generate PDF » button in detail	Invoice OK + lines OK	Reserves no. (if draft), builds PDF, launches	Snackbar + file download

Action	Trigger	Precondition	Effect	Feedback
Cancel invoice	« Cancel invoice » button in detail	Invoice not paid/cancelled	Confirm dialog « Cancel NNN? »	POST up-date_client_invoice_status.php with code='cancelled'; badge turns red Download prompt
Export CSV	« Export CSV » button	Filtered invoices OK	Downloads CSV of displayed invoices (number, client, status, totals, dates, missions)	
Select mission (detail)	Click on mission row in « Associated missions » table	Mission in the invoice	Detail panel displays for just this mission + lines	Linear filter by missionRef, return button « Whole invoice »

6.1.6 User actions — INITIATED INVOICES SECTION

Action	Trigger	Precondition	Effect	Feedback
Load drafts list	Click « Initiated invoices » tab or refresh button	—	GET get_invoice_drafts.php (status=draft)	Spinner; list updates
Resume draft	« Resume » button per draft	Draft present	_restoreFromDraft(draftId), loads lines, switch to Draft section	Panel detach, section switches, table loaded
Delete draft	Trash button + confirm dialog	Draft present	Dialog « Delete draft for [client] [month]? »	POST delete_invoice_draft.php; draft leaves the list

6.1.7 States & edge cases

- **No mission to display:** Empty state (inbox_outlined icon, text « No mission to display »).
- **Missions loading:** Skeleton loader (8 simulated lines).
- **Paid invoice (locked):** « Paid » blue status badge; detail fields read-only (no dropdown, static text).
- **Existing draft for client + month:** « Draft in progress » modal dialog interrupts workflow before loading missions (RM-07).
- **Create invoice without lines:** Button disabled; tooltip explains missing preconditions.
- **No bank account / payment term available:** Optional fields hidden or displayed as static; default text « - ».
- **PDF generation fail:** Error snackbar « Cannot generate PDF ».
- **Viewport width < 860px:** Compact layout (single column, section switcher becomes segmented buttons on top).
- **Mobile (< 600px for mission table toolbar):** Search + line count horizontally at top, pagination + page size at bottom (2 rows).
- **No invoices found:** Empty state (filter_alt_off_outlined icon, text « No result »).
- **Invoice auto-open via route args:** _pendingOpenInvoiceNumber triggers auto-load + auto-select on Invoices tab.

6.1.8 Important business rules

- **Invoice statuses:** Draft → Validated → Paid | Cancelled (non-reversible). Internal code (draft, validated, paid, cancelled) vs FR label.
- **Eligible missions:** Only status 0 (Draft) or 1 (Validated); period filtering = selected month + 12 sliding months max.

- **Locking:** Paid/cancelled invoice = read-only (no status change, no line edit, no date/bank/term edit).
- **Draft autosave:** 800ms after last line edit → POST `save_invoice_draft_lines.php`.
- **Invoice numbering:** Reserved format FAC-YYYYMM-NNN (automatic reservation on creation/PDF gen).
- **Totals:** $HT = \sum(\text{quantity} \times (\text{unitP} \times (1 - \text{discount}/100)))$; $TTC = HT \times (1 + \text{VAT}/100)$.
- **Notes stored:** Invoice keeps « Payment term: XXX » and « Bank account: YYY » in notes field (parse/replace on modify).
- **Invoice pagination:** 25/50/100/250 lines per page; total count in footer; fetch API page 1 with `pageSize=1000` (optimizes client-side pagination).
- **Existing conflict:** If active draft for (client, month) on loading missions → dialog « Resume / Ignore » (RM-07).
- **Stored PDF:** If `invoice.pdfPath` filled, « Open server PDF » button active (`launchUrl` to relative path).

6.1.9 UI dependencies

- **ClientAutocompleteField:** Client search, displays summary (id, name).
- **NumberFormat** (intl): FR currency format (€, 2 dec.) and quantities (0–3 dec.).
- **DataTable:** Lines table (fixed columns, horizontal scroll, sortable via index).
- **showDialog / showDatePicker / showTimePicker:** Core Flutter material dialogs (wrapped via `selectable_dialog.dart` for `TextSelection`).
- **ResponsiveHelper.getSpacing():** Adaptive spacing.
- **BrandFooter:** Persistent footer.
- **Printing:** PDF layout preview (web).
- **UrlLauncher:** Open remote server PDF.
- **CsvExport:** CSV export (via helper stub for web/native).

6.1.10 Notes for the designer (to preserve)

- **Information density:** Draft table = compact (38px header, 40–140px data row); invoices = medium (7 flex columns, paginated display).
- **Colors:** Primary #000091 (dark blue); success accent #15803D (green); error #B91C1C (red); neutral/gray #6B7280–#F3F4F6.
- **Icons:** Material 3 (Google) Icons except custom logo (`assets/logo.png`).
- **Responsive breakpoints:**
 - Desktop ≥ 1300px: wide layout, sidebar expanded, toolbar fit in 1 row.
 - Tablet 860–1300px: sidebar collapsed, section stacked.
 - Mobile < 860px: full stack, SegmentedButton switcher, modals/panels fullscreen or bottom-sheet.
- **Billable text:** Mission status labeling (« Draft », « Validated ») = standard AML.
- **Localization:** FR only (`DateFormat('fr_FR')`, intl strings hard-coded in FR).

6.2 Invoice creation confirmation dialog — `lib/pages/billing_page.dart:5958`

Route / access: `_handleCreateInvoice()`; triggered by « Create and save invoice » button (blue FilledButton).
Business purpose: Confirm with user before final creation (`log_client_billing.php` request); display summary (client, month, line count, totals). **API endpoints:** `log_client_billing.php` (POST, called AFTER confirmation).

6.2.1 Layout & zones

- **Title:** « Create invoice »
- **Content:** 4 summary lines (client, month, line count, total HT, total TTC) — static text, non-editable.
- **Action buttons:**
 - « Cancel » (TextButton, close without action)
 - « Confirm » (FilledButton blue, launches `_createInvoiceFromTable()`)

6.2.2 Data displayed

Field	Source	Format	Notes
Client	<code>_currentClientName</code>	Text	Readonly
Month	<code>_selectedMonth</code>	YYYY-MM via <code>_monthLabel()</code>	Readonly
Lines	<code>_lineEditors.length</code>	Integer	Readonly
Total excl. tax	<code>_currentTotalHt</code>	Currency	Readonly, €2 format
Total incl. tax	<code>_currentTotalTtc</code>	Currency	Readonly, €2 format

6.2.3 User actions

Action	Trigger	Effect	Feedback
Confirm creation	« Confirm » button	<code>Pop(true)</code> , call <code>_createInvoiceFromTable()</code>	Dialog closes, « Creating... » spinner then result snackbar
Cancel	« Cancel » button or Esc	<code>Pop(false)</code>	Dialog closes without action

6.2.4 States & edge cases

- **Missing preconditions:** Dialog never displayed; snackbar « Please select... » on create click.
- **Creation in progress:** Create button becomes disabled, spinner appears; feedback in snackbar if success/error.

6.2.5 Important business rules

- **Validation before dialog:** `_validateInvoiceActionRequirements(forCreation: true)` — client, month, lines, no existing invoice conflict.
- **Reserved number:** Once confirmed, `reserveNextInvoiceNumber()` is called (transactional, incremental per YYYYMM).
- **PDF generated:** `_buildInvoicePdfFile()` executed immediately after, bytes sent to API.

6.3 Invoice cancellation confirmation dialog — `lib/pages/billing_page.dart:310`

Route / access: `_cancelInvoice(invoice)`; triggered by « Cancel invoice » button or context menu action. **Business purpose:** Confirm before transition to « Cancelled » status (irreversible, affects accounting). **API endpoints:** `update_client_invoice_status.php` (POST, called AFTER confirmation).

6.3.1 Layout & zones

- **Title:** « Cancel invoice »
- **Content:** Text « Confirm cancellation of invoice [INV_NUMBER]? »
- **Action buttons:**
 - « Close » (`TextButton`)
 - « Cancel invoice » (`FilledButton` blue)

6.3.2 Data displayed

Field	Source	Format	Notes
Invoice no.	<code>invoice.invoiceNumber</code>	Text	Interpolated in text

6.3.3 User actions

Action	Trigger	Effect	Feedback
Confirm cancellation	« Cancel invoice » button	Pop(true), call <code>_changeInvoiceStatus(invoice, 'Cancelled')</code>	Dialog closes, status update (spinner badge), snackbar or error
Close without action	« Close » button or Esc	Pop(false)	Dialog closes, invoice unchanged

6.3.4 Important business rules

- **Precondition:** `_canCancelInvoice(invoice) = status ≠ paid && ≠ cancelled`.
- **Target status:** Code='cancelled', label='Cancelled', visible red badge.
- **Optimistic update:** UI change immediate, API confirm async; revert if error.

6.4 Invoice date/time selection dialog — `lib/pages/billing_page.dart:4191`

Route / access: `_pickInvoiceDate(invoice, current)`; triggered by click on date field in invoice detail (invoice not locked). **Business purpose:** Allow modification of creation/billing date (e.g.: backdating for adjustment). **API endpoints:** `update_invoice_date.php` (POST, called AFTER selection).

6.4.1 Layout & zones

- **Step 1: Date picker:** Flutter material calendar (`firstDate = year-10`, `lastDate = year+5`, `initialDate = current` or `now()`).
- **Step 2: Time picker:** Hour/minute selection (24h format).

6.4.2 User actions

Action	Trigger	Effect	Feedback
Select date	Click day in calendar	Move to time picker	Calendar closes, time picker displays
Select time	Click hour/minute	Pop with complete <code>DateTime</code>	Time picker closes, POST <code>update_invoice_date.php</code> ; snackbar if success
Cancel	Click Cancel or Esc	Pop without change	Dialog(s) close, date unchanged
Revert if error	API error	Revert to previous value, error snackbar	Invoice refresh with old date

6.4.3 Important business rules

- **Time preservation:** If `current != null`, preserves existing time; otherwise uses `TimeOfDay.now()`.
- **Allowed range:** `now.year ± 10` (broad range for accounting flexibility).
- **Optimistic update:** Displays new date immediately, API confirm async.

6.5 « Draft in progress » dialog (Resume/Ignore Draft) — lib/pages/billing_page

Route / access: `_checkForExistingDraft()`; triggered automatically when loading missions if an active draft exists for (client, month). **Business purpose:** RM-07 workflow — propose resuming an existing draft rather than duplicating (additive approach). **API endpoints:** `get_invoice_drafts.php` (POST, to check existence); `get_invoice_draft_lines.php` (POST, if resume).

6.5.1 Layout & zones

- **Title:** « Draft in progress »
- **Content:**
 - Text « A draft exists for [CLIENT] »
 - Draft details: Month, Total HT
 - Question « What do you want to do? »
- **Action buttons:**
 - « Ignore » (TextButton)
 - « Resume » (FilledButton blue)
- **Behavior:** `barrierDismissible: false` (forced modal).

6.5.2 Data displayed

Field	Source	Format	Notes
Client	<code>draft.clientName</code> \\	Text	Client name of existing draft
Month	<code>draft.month</code>	YYYY-MM	Period stored in draft
Total excl. tax	<code>draft.totalHt</code>	Currency, 2 dec.	Current total in draft

6.5.3 User actions

Action	Trigger	Effect	Feedback
Resume	« Resume » button	<code>Pop('resume')</code> , call <code>_restoreFromDraft(draft)</code>	Dialog closes, « Loading... » spinner, table populates with draft lines
Ignore	« Ignore » button	<code>Pop('ignore')</code>	Dialog closes, draft remains in DB, raw missions displayed instead

6.5.4 Important business rules

- **Detection:** `get_invoice_drafts.php` request filtered by `client_name + month = 'YYYY-MM'`; if ≥ 1 result → dialog appears.
- **Restoration:** `_restoreFromDraft()` loads lines (`get_invoice_draft_lines.php`), populates `_lineEditors`, `_draftId` + `_draftKey` reset.
- **No error:** If API error during detection, dialog does not display (graceful degrade).

6.6 Draft deletion confirmation dialog — lib/pages/billing_page.dart:5917

Route / access: `_buildDraftsList()` — triggered by trash icon click in drafts `ListView`. **Business purpose:** Confirm before final deletion of a draft (`delete_invoice_draft.php` call). **API endpoints:** `delete_invoice_draft.php` (POST).

6.6.1 Layout & zones

- **Title:** « Delete draft? »
- **Content:** « The draft for [CLIENT] ([MONTH]) will be deleted permanently. »
- **Action buttons:**
 - « Cancel » (TextButton)
 - « Delete » (FilledButton red #B91C1C)

6.6.2 Data displayed

Field	Source	Format	Notes
Client	<code>draft.clientName</code> \\ 'this client'	Text	Fallback if null
Month	<code>draft.month</code>	YYYY-MM	Draft period

6.6.3 User actions

Action	Trigger	Effect	Feedback
Confirm deletion	« Delete » button (red)	Pop(true), POST <code>delete_invoice_draft.php</code>	Dialog closes, draft leaves list, snackbar confirmed
Cancel	« Cancel » button or Esc	Pop(false)	Dialog closes, draft preserved

6.6.4 Important business rules

- **Precondition:** Draft status = 'draft' (non finalized); otherwise trash button disabled.
- **Cascade:** Draft deletion → associated `invoice_draft_lines` deleted in CASCADE (SQL).
- **Revert on error:** If delete fails, snackbar « Error: ... »; draft stays in list.

6.7 Billing Line Editor Panel — `lib/pages/billing_page.dart`

Route / access: `_buildLineEditorPanel(spacing)`; triggered by click on a table line or via `_openLineEditor(index)`.
Business purpose: Detailed editing interface for a line (description, quantity, price, VAT, notes); lateral access (right side panel). **API endpoints:** None (in-memory editing; auto-persist via draft autosave).

6.7.1 Layout & zones

- **Header:** Title « Line editing »; close button (×).
- **Editing fields:**
 - Description (TextFormField, 3 lines, editable)
 - Quantity & Unit price excl. tax (Row, 2 TextFormFields, numeric)
 - VAT (DropDownButtonFormField, % selection)
 - Notes (TextFormField, 2 lines, optional)
 - **Display:** Line Total HT + Linked mission (static texts)
- **Action buttons:**
 - « Reset line » (OutlinedButton)
 - « Reset all » (FilledButton, optional if >1 line)

6.7.2 Data displayed

Field	Source	Format	Type	Notes
Description	line.designation	Text, 3 lines max	Editable	Can be multi-line
Quantity	line.quantity	Decimal, 3 dec.	Editable	Number field
Unit price excl. tax	line.unitPrice	Decimal, 2 dec.	Editable	Number field
Unit price incl. tax	line.unitPriceTtc	Currency	Read-only	Auto-computed
VAT	line.tvaRate	Dropdown (%; e.g.: 0, 5.5, 20)	Editable	Selection among _tvaRates
Notes	line.notes	Text, 2 lines	Editable	Optional
Total excl. tax	line.totalHt	Currency	Read-only	Computed = (qty × (price × (1 - discount%)))
Mission	editor.missionRef	Text	Read-only	Linked mission ref (static)

6.7.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Edit description	Typing in TextField	Panel open	Updates _lineEditors[index].currentLine.designation	Text reflected immediately
Edit quantity	Typing in TextField	Panel open	Updates _lineEditors[index].currentLine.quantity	Input validated (>0); totals auto-update
Edit unit price excl. tax	Typing in TextField	Panel open	Updates unitPrice, recalc totalHt/totalTtc	Input validated (≥0)
Select VAT	Click Dropdown	Panel open	Updates tvaRate, recalc unitPriceTtc/totalTtc	Dropdown toggles, totals recomputed
Edit notes	Typing in TextField	Panel open	Updates notes (null if empty)	Text reflected
Reset line	« Reset line » button	Panel open, line modified	Reset _lineEditors[index] to original values (_EditableInvoiceLine.reset())	Form controllers reload; values revert
Reset all	« Reset all » button	Panel open, >1 line, any modified	Reset all _lineEditors	All controllers reload
Close	Click × or ESC	—	Panel closes, _showLinePanel=false, _selectedLineIndex=null	Panel slide-out, table refresh
Auto-persist	After editing, 800ms delay	Any change	Scheduler autosave via _draftSaveTimer	Transient « Saving... » badge

6.7.4 States & edge cases

- **No line selected:** Panel does not appear.
- **Line deleted during editing:** Panel remains visible, controllers orphan (no effect if save attempted).
- **Modification then closed:** Changes kept in memory _lineEditors; autosave via timer.

6.7.5 UI dependencies

- **TextEditingController** × 4 (description, quantity, unitPrice, notes).
- **DropDownButtonFormField** for VAT.
- **NumberFormat** for displaying/parsing quantities/currencies.

6.8 Invoice Details Panel — lib/pages/billing_page.dart

Route / access: `_buildInvoiceDetailsPanel(spacing)`; triggered by click on an invoice in the table (or route args with invoiceNumber). **Business purpose:** Display/modify invoice header (status, dates, settings), visualization of lines by mission, line editing if invoice not locked. **API endpoints:** - `get_client_invoice_lines.php` (POST, fetch lines) - `update_client_invoice_lines.php` (POST, edit lines per mission) - `update_client_invoice_status.php` (POST, status) - `update_invoice_bank_account.php` (POST, bank account) - `update_invoice_payment_term.php` (POST, payment term) - `update_invoice_date.php` (POST, date)

6.8.1 Layout & zones

- **Header:** Invoice no. (blue, large), status badge, client name (gray, medium size), chips (Month, Creation date, mission scope if selection).
- **Section 1 « Overview »:** Month, Creation date (editable), TotalHT (highlighted), Status (dropdown or display).
- **Section 2 « Settings »:** Payment term (dropdown or static), Bank account (dropdown or static).
- **Section 3 « Associated missions »:** Compact table of missions (ref, description, unit price, Qty, Total HT); click on mission for detail-view.
- **Section 4 « [MissionRef] detail »** (if mission selected): Editable list of mission lines + Reset/Save buttons.
- **Footer:** « Generate PDF » buttons (full-width), « Cancel invoice » (full-width, red).
- **Close:** × button at top-right.

6.8.2 Data displayed (OVERVIEW)

Field	Source	Format	Type	Notes
Invoice no.	<code>invoice.invoiceNumber</code>	Text, link-blue	Read-only	Large title
Client	<code>invoice.clientName</code>	Text	Read-only	Medium gray
Status	<code>invoice.statusCode/statusBadge</code>	Color badge	Editable (dropdown)	Read-only if locked
Month	<code>invoice.periodMonth</code>	YYYY-MM	Read-only	Chip
Creation date	<code>invoice.createdAt</code> <code>/ billedAt</code>	dd/MM/yyyy HH:mm	Editable (picker)	Read-only if locked
Total excl. tax	Sum of lines or <code>invoice.invoiceTotalHt</code>	Currency	Read-only	Highlighted, computed

6.8.3 Data displayed (ASSOCIATED MISSIONS)

Field	Source	Format	Type	Notes
Mission ref	<code>line.missionRef</code>	Text	Read-only	Clickable → detail view
Description	First line description	Text	Read-only	Max 2 lines overflow
Unit price excl. tax	If 1 single line = <code>unitPrice</code>	Currency	Read-only	Otherwise « - » (multi-line)
Qty	Sum of group quantities	Decimal	Read-only	Qty format

Field	Source	Format	Type	Notes
Total excl. tax	Sum of group totalHt	Currency	Read-only	Highlight

6.8.4 Data displayed (MISSION DETAIL — EDITING)

Field	Source	Format	Type	Notes
Description	line.designation	Text, editable, multi	Editable	Container per line
VAT	line.tvaRate	Dropdown %	Editable	Locked if invoice.locked
Unit price excl. tax	line.unitPrice	Number	Editable	Locked if invoice.locked
Qty	line.quantity	Number, 3 dec.	Editable	Locked if invoice.locked
Unit price incl. tax	line.unitPriceTtc	Currency	Read-only	Calculated
Discount %	line.discount	Number	Editable	0–100
Notes	line.notes	Text, editable	Editable	Optional
Total excl. tax	line.totalHt	Currency	Read-only	Calculated
Total incl. tax	line.totalTtc	Currency	Read-only	Calculated

6.8.5 User actions

Action	Trigger	Precondition	Effect	Feedback
Load lines	Panel open	—	POST get_client_invoice_lines.php	Centered spinner, populates
Change status	Status dropdown	Invoice not locked	Optimistic update, POST up-date_client_invoice_status.php	Badge toggles; spinner overlay; spinner if error
Modify creation date	Click on date (calendar icon)	Invoice not locked	showDatePicker + showTimePicker → POST up-date_invoice_date.php	Picker dialog opens; date update optimistic
Modify bank account	« Bank account » dropdown	Invoice not locked + accounts available	Optimistic update, POST up-date_invoice_bank_account.php	Dropdown toggle; spinner; notes update
Modify payment term	« Payment term » dropdown	Invoice not locked + terms loaded	Optimistic update, POST up-date_invoice_payment_term.php	Dropdown toggle; spinner; notes update
Select mission (detail-view)	Click mission row	Mission exists	Panel rewrites with mission lines; « Whole invoice » return button appears	Section 4 populates, Section 3 hidden
Return to whole invoice	Click « Whole invoice » button	Mission detail-view active	Panel returns to full view (all missions)	Section 4 hidden, Section 3 visible
Edit line (mission detail)	Typing in line container	Invoice not locked + mission detail active	Updates _selectedInvoiceLineEditors[index].currentLine	Fields reflect edits, totals recomputed
Reset lines	« Reset » button	Mission detail active, any modified	Reset _selectedInvoiceLineEditors to originals	All fields reload

Action	Trigger	Precondition	Effect	Feedback
Save lines	« Save lines » button	Invoice not locked, changes detected	POST <code>update_client_invoice_lines.php</code>	« Saving... » spinner; success/error snackbar; panel refresh
Generate PDF	« Generate PDF » button	Lines OK	Reserves no. if draft, builds PDF, launches download	Spinner; confirmation snackbar
Cancel invoice	« Cancel invoice » button (red)	Invoice not paid/cancelled	Dialog confirm, POST <code>update_client_invoice_status.php</code> with <code>code='cancelled'</code>	Status toggles to « Cancelled » (red); panel closes or remains read-only
Close detail	Click ×	—	Panel closes, <code>_selectedInvoice=null</code> , <code>_selectedInvoiceLines=null</code>	Panel slide-out (right align)

6.8.6 States & edge cases

- **Paid invoice (locked):** All fields read-only (no dropdown, static text); persistent « Paid » badge; Cancel button disabled.
- **No associated mission:** Text « No mission associated with this invoice. »
- **Initially draft invoice:** Full editing allowed (all fields, active buttons).
- **Line loading in progress:** Centered spinner section 3/4.
- **No bank accounts / terms available:** Dropdowns hidden or displayed as static (read-only).
- **Mobile (< 860px):** Panel fullscreen overlay (no side panel); close button top-right; vertical scroll.
- **Desktop (≥ 860px):** Right side panel, max-width 640px, slide-in animation from right.

6.8.7 Important business rules

- **Locking:** `invoiceIsLocked(pdo, invoiceNumber)` in PHP; status ∈ {paid, payée, payee, reglee, cancelled, annulee} = locked.
- **Notes stored:** `Invoice.notes` contains « Payment term: XXX » + « Bank account: YYY » (parse/replace).
- **Mission totals:** $\text{Sum}(\text{quantity} \times \text{unitPrice} \times (1 - \text{discount}/100))$ per mission group.
- **Optimistic modification:** UI change immediate, API confirm async, revert if error.
- **Edit lines per mission:** `update_client_invoice_lines.php` takes `invoiceNumber` + `missionRef` + line list.

6.8.8 UI dependencies

- **Spinner** (`CircularProgressIndicator`) if `_loadingInvoiceLinesPanel`.
- **TextEditingController** × N for lines (re-dynamized per editing).
- **DropDownButtonFormField** for status, bank account, payment term.
- **DatePicker / TimePicker** (`showDatePicker`, `showTimePicker`).
- **Responsive layout:** `ConstrainedBox`, `Expanded`, scrollable `ListView`.

6.9 Fullscreen Lines Table Overlay — `lib/pages/billing_page.dart`

Route / access: `_buildFullscreenOverlay(spacing)`; triggered by click on « See in fullscreen » button (fullscreen_outlined icon) in Draft section. **Business purpose:** Maximized display of the billing lines table (desktop scenario with little space). **API endpoints:** None (same data as standard table, local editing).

6.9.1 Layout & zones

- **Header:** fullscreen icon (info), title « Fullscreen view of lines », line count, close button (×).
- **Toolbar:** Search filter + page size + line count + pagination.
- **Body:** DataTable identical to standard view (ref/description/VAT/unit price/etc. columns, horizontal/vertical scroll).
- **Footer:** HT/TTC totals right-aligned.
- **Background:** Semi-transparent modal overlay (black 54%).

6.9.2 User actions

Action	Trigger	Effect	Feedback
Filter lines	Search TextField	Client-side filter on _missionTableSearch	Table reflects query
Paginate	Pagination buttons	Change _missionTablePage	Table scroll to top, new range display
Edit line	Click cell	Inline editing identical (qty/price/VAT/notes)	Cell editable, auto recalc
Close fullscreen	× button	Pop overlay, _isTableFullscreen=false	Return to standard view
ESC	ESC key	Same as close	Overlay disappears

6.9.3 States & edge cases

- **No line:** Overlay does not appear.
- **Close without saving:** Changes remain in memory, autosave via timer.

6.10 « Initiated invoices » section (Preparations List) — lib/pages/billing_page.d

Route / access: « Initiated invoices » tab (mobile SegmentedButton or desktop sidebar item). **Business purpose:** List drafts (preparations) saved but not yet finalized; allow resume or delete. **API endpoints:** - get_invoice_drafts.php (POST) - delete_invoice_draft.php (POST)

6.10.1 Layout & zones

- **Title:** « Initiated invoices »
- **Subtitle:** « Invoices saved but not yet finalized. »
- **Body:** Card with ListView.separated of drafts (or empty state / error state).
- **Per item:**
 - Blue description icon
 - Title: Client name
 - Subtitle: « Month: [MONTH] • Total: [HT] € »
 - Trailing Row: « Resume » button (play icon) + delete icon (trash, red)

6.10.2 Data displayed

Field	Source	Format	Notes
Client	draft.clientName \\\ '(unknown client)'	Bold text	Title
Month	draft.month	YYYY-MM	Subtitle
Total excl. tax	draft.totalHt	Currency, 2 dec.	Subtitle

Field	Source	Format	Notes
Status	<code>draft.status</code>	Enum 'draft'/'finalized' (hidden)	Logic: finalized unavailable

6.10.3 User actions

Action	Trigger	Precondition	Effect	Feedback
Load list	Switch to « Initiated invoices » tab	—	GET <code>get_invoice_drafts.php</code>	Centered spinner, list populates
Refresh	Click refresh icon (top-right)	—	Reload <code>get_invoice_drafts.php</code>	List update, transient spinner
Resume draft	« Resume » button per item	Draft present	<code>Pop() → _restore-FromDraft(draft);</code> switch to Draft section	Dialog closes, section switches, table loaded
Delete draft	Click trash icon per item	Draft present	Confirm dialog « Delete draft for [client] ([month])? »	Confirm → POST <code>delete_invoice_draft.php</code> ; draft leaves list

6.10.4 States & edge cases

- **Loading:** Centered spinner.
- **API error:** Error state (red `error_outline` icon, error text, « Retry » button).
- **No draft:** Empty state (gray `folder_copy_outlined` icon, text « No draft in progress. »).

Chapter 7

Appendix A — Pages × Endpoints Matrix

Page / Dialog	Consumed API endpoints
Login	api/login.php
Home / Dashboard	api/get_missions_datatable.php, api/get_clients.php
Interpreter list	api/get_interpretes.php, api/delete_interprete.php, api/get_languages.php
Add/edit interpreter dialog	api/add_interprete.php, api/update_interprete.php, api/get_countries.php, api/get_departments.php, api/get_languages.php
Missions per interpreter page	api/get_missions_by_interpreter.php, api/get_tab_mission_par_interpreters.php
Missions table	api/get_missions_datatable.php, api/add_mission_interpreter.php, api/update_mission_interpreter.php, api/delete_mission_interpreter.php, api/create_quote_from_mission.php
Companies / contacts management	api/get_clients.php, api/add_client.php, api/update_client.php, api/delete_client.php, api/get_contacts.php, api/add_contact.php, api/update_contact.php, api/delete_contact.php
Billing page	api/get_client_invoices.php, api/get_client_invoice_lines.php, api/get_invoice_drafts.php, api/get_invoice_draft_lines.php, api/save_invoice_draft.php, api/save_invoice_draft_lines.php, api/update_client_invoice_lines.php, api/update_client_invoice_status.php, api/update_invoice_bank_account.php, api/update_invoice_date.php, api/update_invoice_payment_term.php, api/reserve_client_invoice_number.php, api/delete_invoice_draft.php, api/convert_quote_to_invoice.php, api/log_client_billing.php, api/get_client_payment_terms.php, api/get_company_bank_accounts.php

Page / Dialog	Consumed API endpoints
Quote editing page	api/get_quote.php, api/update_quote.php, api/create_quote_from_mission.php, api/convert_quote_to_invoice.php,
Company settings	api/get_company_info.php api/get_company_info.php, api/update_company_info.php, api/get_company_bank_accounts.php
Admin / users page	api/admin/* (add_user, update_user, list_users...)
Export page	api/admin/exportAll, api/admin/exportMissions

Chapter 8

Appendix B — Business glossary (FR ↔ EN)

Français	Anglais	Definition
Mission	Mission	Scheduled interpretation service for a requester company, on a given date, for one or more languages.
Interprète	Interpreter	Individual provider performing the mission. Can be internal or external.
Demandeur / Société	Requester / Company	Client (legal entity) ordering the mission.
Contact	Contact	Natural person attached to a requester company.
Devis	Quote	Preliminary commercial document, modifiable as long as status ≠ {draft, sent}.
Facture	Invoice	Fiscal document locked upon validation, numbered FAC-YYYYMM-NNN.
Brouillon de facture	Invoice draft	Invoice not yet numbered, freely modifiable and deletable.
Ligne de facture / devis	Invoice / Quote line	Detailed service (description, unit price excl. tax, quantity, VAT, discount).
Terme de paiement	Payment term	Contractual delay (e.g.: 30 days end of month), source 11x_c_payment_term.
Compte bancaire société	Company bank account	Bank details (RIB/IBAN) displayed in the invoice footer, managed in Company settings.
Statut mission	Mission status	Enumeration: to plan, planned, completed, billed, cancelled... (see mission-status-semantics).
Statut devis	Quote status	draft → sent → accepted/rejected/expired → accepted_converted.
Statut facture	Invoice status	draft → validated → paid / cancelled.
Rôle admin	Admin role	Access right to configuration, users and exports.
Rôle gestionnaire	Manager role	Right to manage missions, clients, contacts, invoices.

Français	Anglais	Definition
Rôle interprète	Interpreter role	Restricted view on own missions.

Chapter 9

Appendix C — Screenshots

Screenshots are stored in [docs/figma-handoff/screenshots/](#) and referenced in each sheet by their slug `<!-- screenshot-slug: ... -->`. They are provided as **visual reference only**: the Figma designer is free to entirely rethink the layout, provided that the **data displayed**, **user actions**, **states** and **business rules** described in the corresponding sheet are preserved.